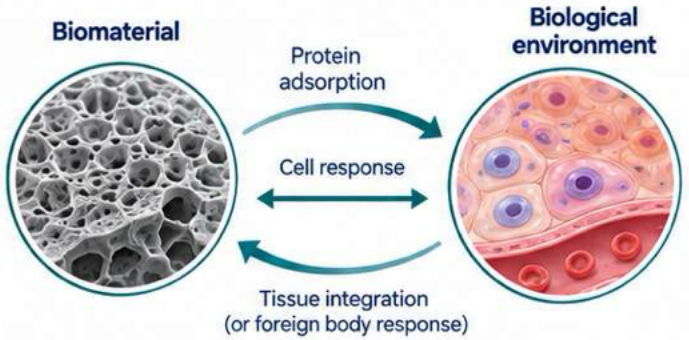
Biomedical Materials

Week 1: Introduction Recap

Week 2 lecture slides
Undergraduate course

What Is a Biomaterial?

A biomaterial is a material designed to interact with biological systems for a medical purpose such as diagnosis, treatment, repair, or replacement of tissue or function.



Common roles

- Replace damaged tissue
- Support healing or regeneration
- Restore function
- Deliver drugs or signals
- Assist diagnosis or monitoring

Examples

Hip and knee implants

Load-bearing joint replacement

Dental implants

Tooth replacement and restoration

Cardiovascular stents

Keep vessels open and restore blood flow

Contact lenses

Vision correction and drug delivery

Sutures or catheters

Support healing and deliver therapies

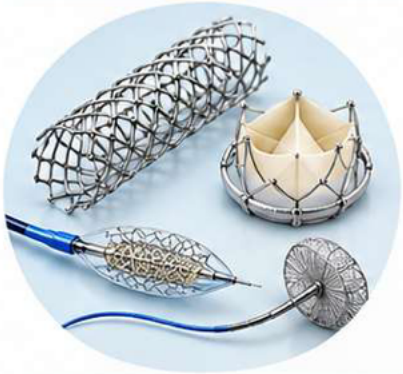
Why Biomedical Materials Matter

Orthopedics

Bone plates, screws, hip stems, and knee components restore mechanical support and mobility.

Dentistry

Implants, crowns, orthodontic wires, and fillings help restore oral function.



Cardiovascular

Stents, grafts, and heart-valve components support blood flow and cardiac function.

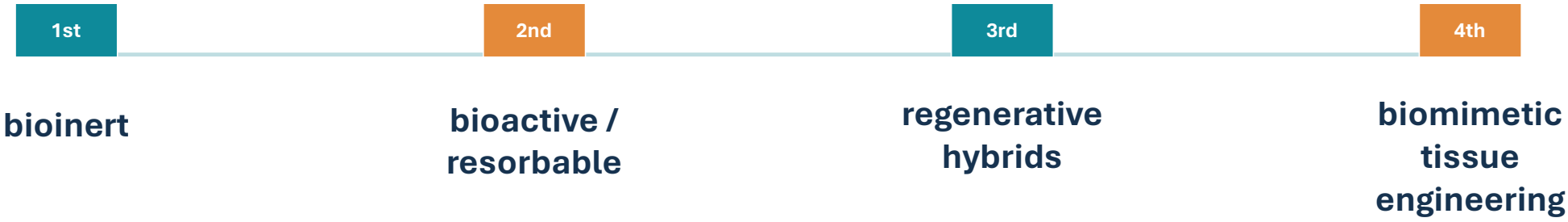
Ophthalmic / Soft Tissue

Contact lenses, intraocular lenses, catheters, and sutures support vision and soft-tissue care.

Biomedical materials improve quality of life by restoring function and supporting treatment.

Four generations of biomaterials

A teaching framework for how material goals evolved



Major Classes of Biomedical Materials

Metals

Strong and tough, widely used for load-bearing implants.

Examples:
stainless steel, Co-Cr alloys, titanium alloys

Ceramics

Hard, wear-resistant, and often bioactive or bioinert.

Examples:
alumina, zirconia, hydroxyapatite

Polymers

Lightweight and flexible, useful in both permanent and temporary devices.

Examples:
UHMWPE, PMMA, silicone

Composites

Combine properties of two or more materials.

Examples:
fiber-reinforced and polymer-ceramic systems

Natural / Biobased

Derived from biological sources and often used in regenerative applications.

Examples:
collagen, chitosan

Not all biomaterials are load-bearing

Properties that matter

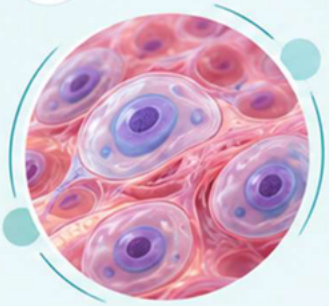
1 Mechanical

strength, stiffness, fatigue, wear

2 Chemical

corrosion, degradation, water uptake

3 Biological



biocompatibility, immune response, infection

4 Processing

manufacturing, surface finishing, sterilization

Mechanical Properties of Biomaterials

Properties must match the device function and loading environment

Strength
Resists applied force.

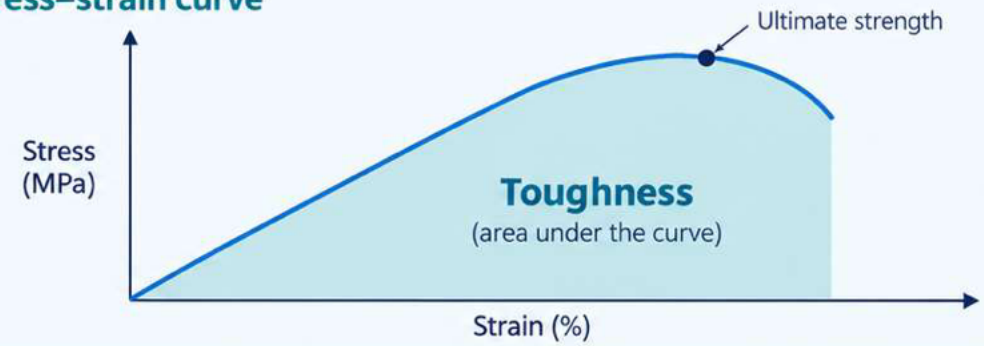
Toughness
Absorbs energy before fracture.

Elastic modulus
Controls stiffness and load transfer.

Wear resistance
Limits debris generation.

Hardness
Resists surface indentation and scratching.

Stress–strain curve



Wear, debris generation, and biological response

Two materials rubbing in a biological environment

Wear debris
(metal and/or polymer particles)

Possible inflammatory response

Metal debris may release ions e.g., Co^{2+}, Cr^{3+}, Ni^{2+}.

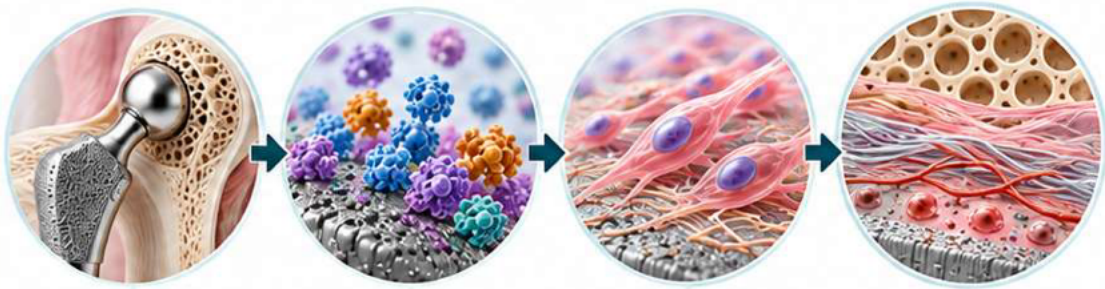
Polymer debris can trigger foreign body / chronic inflammatory reactions.

Biocompatibility and Biological Performance

Biocompatibility

- Ability of a material to perform its intended function in the body without causing unacceptable harm.
- Depends on material composition, surface, shape, and location in the body.
- The best material is not always the strongest; it must also be biologically acceptable.
- Biocompatibility is application-specific.

Biological performance



Key idea: material success depends on both engineering performance and host response.

Possible Biological Outcomes

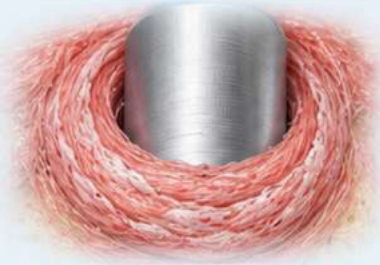
Integration, encapsulation, degradation, or failure

✓ Integration

desired tissue bonding or stable fixation

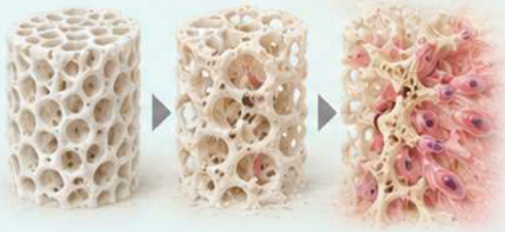
Caps Encapsulation

fibrous tissue isolates device from host



↓ Resorption

biodegradable material is replaced or cleared



× Failure

infection, loosening, wear, fracture, thrombosis, toxicity

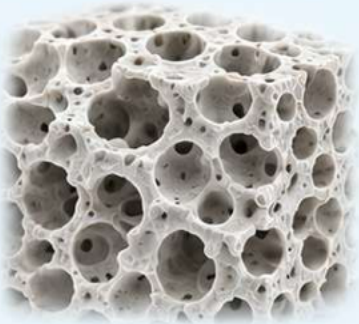
Biological performance is judged against the intended function and lifetime.

Surface Properties of Biomaterials

The surface often controls early biological response

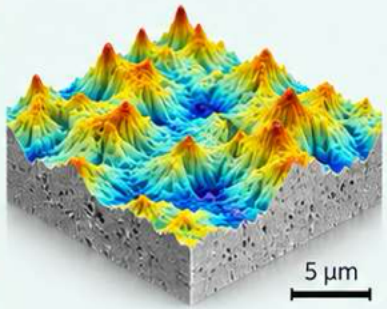
1 Porosity

Voids and cavities accessible to molecules.



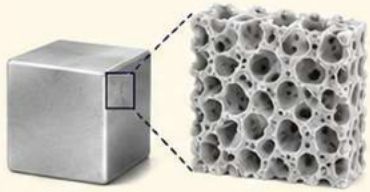
2 Surface roughness

Topography at micro- and nano-scale.



3 Surface area

Exposed surface available for interaction.



Smooth surface (lower area)

Rough/porous surface (higher area)

4 Wettability

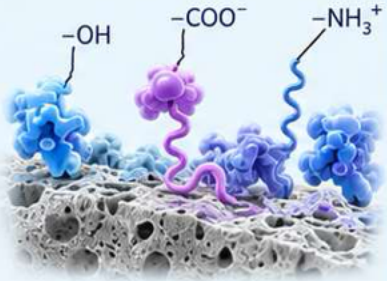
Tendency of fluid to spread or adhere.

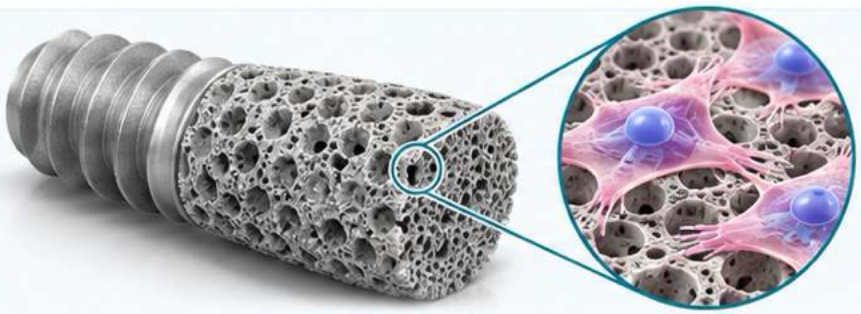
More hydrophobic (higher contact angle)

More hydrophilic (lower contact angle)

5 Surface chemical structure

Functional groups and charge.





These properties affect **protein adsorption, osteoblast proliferation, and tissue reaction.**

Sterility and Infection: Introductory View

A biomaterial must be safe before it reaches the patient

- Sterilization removes or inactivates microorganisms before device use.
- Methods must not damage the material or change its function.
- Bacterial adhesion and biofilm formation can cause serious implant failure.
- Surface design and surgical handling both influence infection risk.

Ster **Common sterilization routes**

steam, ethylene oxide, radiation, plasma-based methods

Steam (autoclave) | Ethylene oxide | Radiation | Plasma-based methods

Poly **Materials issue**

Some polymers degrade or age after heat or radiation exposure.

Bioinert, Bioactive, and Biodegradable Materials

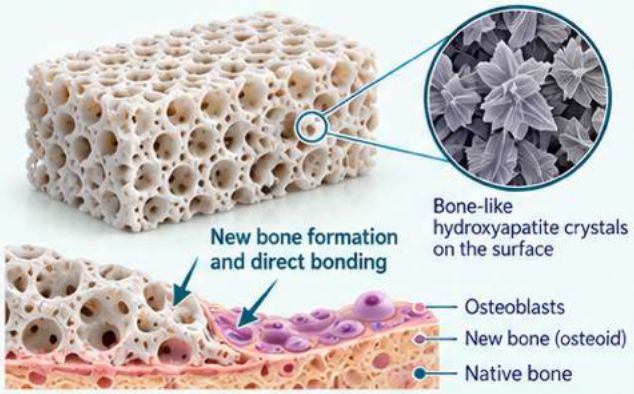
Bioinert

Causes minimal interaction with surrounding tissue.

Examples:
alumina, zirconia, titanium alloys, medical-grade stainless steel

Bioactive

Interacts positively with tissue and may bond to bone or stimulate a response.



Examples:
hydroxyapatite, bioactive glass, bioactive ceramics

Biodegradable

Gradually breaks down in the body over time.

Initial scaffold
Provides mechanical support

Progressive degradation
Maintains structural integrity while tissue regenerates

Resorbed
Replaced by new tissue

Examples:
PLA, PGA, PLGA, magnesium-based materials, biodegradable polymers

Material choice depends on the application and the desired interaction with the body.

Introductory material-selection criteria

Move from vocabulary to engineering decision-making.

Mechanical	Chemical	Biological	Processing	Economics

Function is the starting point

What biomaterials are expected to do inside or near the body

R **Replace**

damaged tissue or missing structure

S **Support**

healing, fixation, or regeneration

F **Restore**

mechanical, optical, or electrical function

D **Deliver**

drugs, signals, or therapeutic agents

M **Monitor**

diagnostic information or physiological state

Design begins with the desired function and the local biological environment.

The design triad

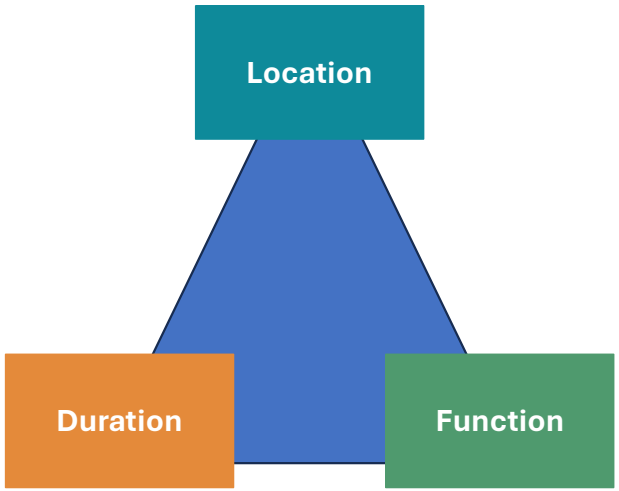
The same material can be good or poor depending on these three variables

Location determines pH, fluid chemistry, cells, proteins, salts and bacteria.

Duration determines exposure time, degradation risk and testing burden.

Function determines loads, motion, transport, sensing or tissue interaction.

Material selection changes whenever one part of the triad changes.



If a device moves from bone to blood, which part of the design triad changes most strongly?

Biomedical Materials

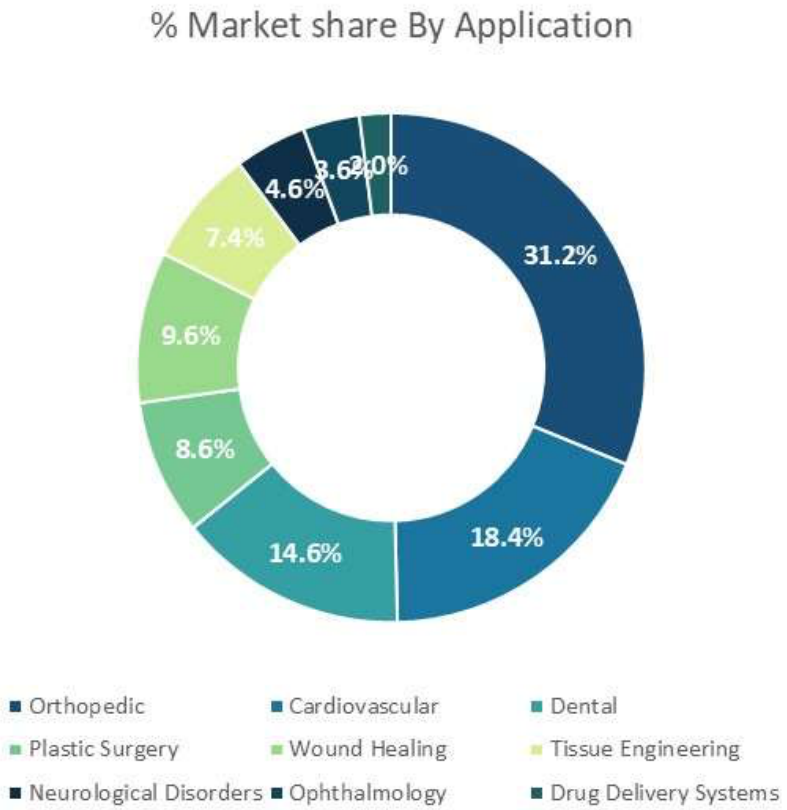
Week 2: Bone as a Material

Structure, composition and properties

Week 2 lecture slides
Undergraduate course

Biomaterials Market, By Application, 2026

Total Market Size:
USD 241.55 Bn



Source: Primary Research, Desk Research, Paid subscriptions, CMI Data Repository

COHERENT MARKET INSIGHTS

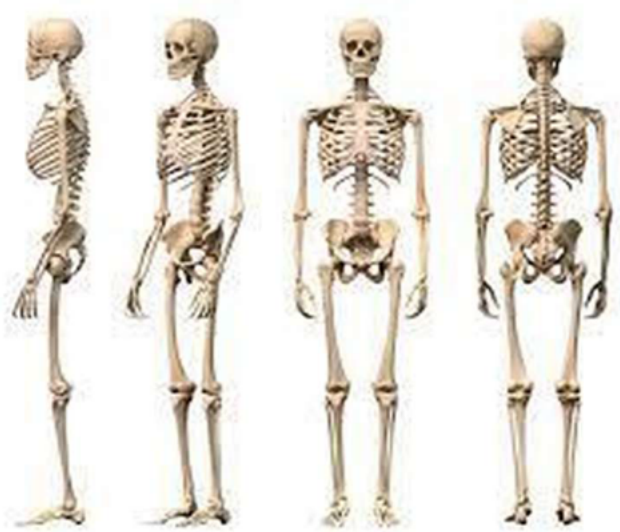
31.20%

Orthopedic Application - Estimated Market Revenue Share, 2026

Biomaterials Market

Introduction

- A living bone is an extremely **dynamic** tissue that is continually **modelled** and **remodelled** by the forces acting on it.
- Bone is **living tissue** that makes up the body's skeleton
- The **human skeleton** is made up of between 206 and 210 bones.
- Bones are categorized according to various shapes:



Function of skeleton

- The skeleton consists of approximately 20% of total body weight.
- Bone tissue performs many functions:
  - Support
  - Attachment sites
  - Leverage
  - Protection
  - Storage
  - Blood cell formation.

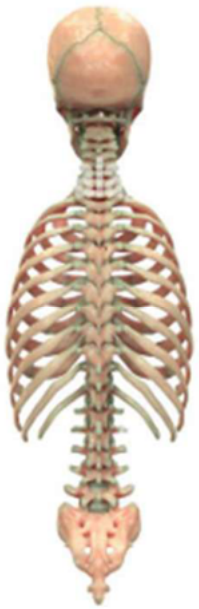
Function of skeleton

- **Support:** Structural support and can maintain a posture while accommodating large external forces.
- **Attachment sites:** Provide sites of attachment for **tendons**, **muscles**, and **ligaments**, allowing for the generation of movement.
- **Leverage:** Provides the **levers** and **axes** of rotation about which the muscular system generates the movements.
- **Protection:** Provides protection of **vital organs.**
- **Storage:** Stores fats and minerals (*calcium and phosphate)
- **Blood cell formation:** Haematopoiesis

Anatomical classification of bones

- Skeletal system has two main parts:
  - The axial (skull, spine, ribs, and sternum)
  - Appendicular (shoulder, pelvic girdles, arm, and legs)

Appendicular Skeleton



Axial Skeleton

Anatomical classification of bones

- Long: offer the body support, the interconnected set of levers, and linkages that allow us to move.

  e.g. humerus, femur, tibia, radius ulna fibula

- Short: role in shock absorption and the transmission of forces.

  e.g. phalanges, metatarsals patella

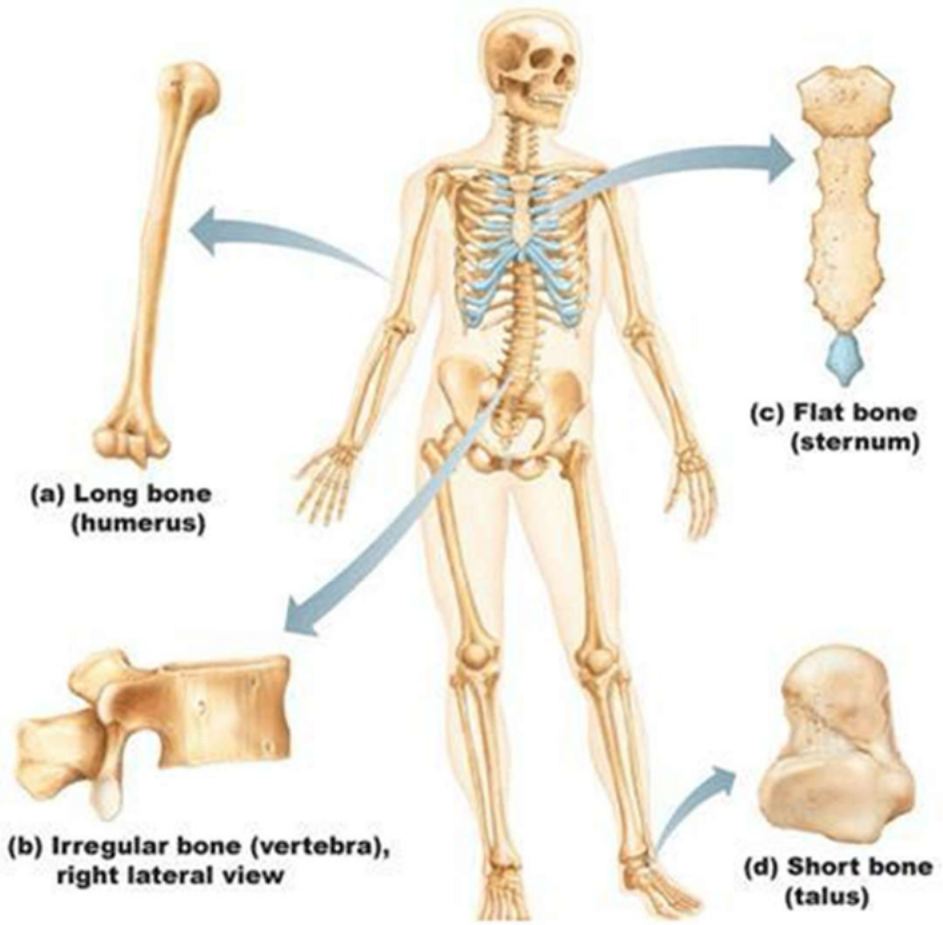
- Flat: protect internal structures and offer broad surfaces for muscular attachment.

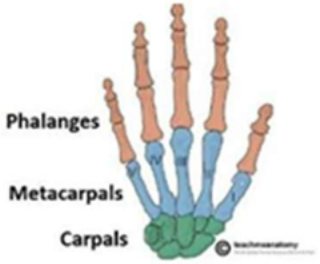
  e.g. scapula, illium, sternum, skull

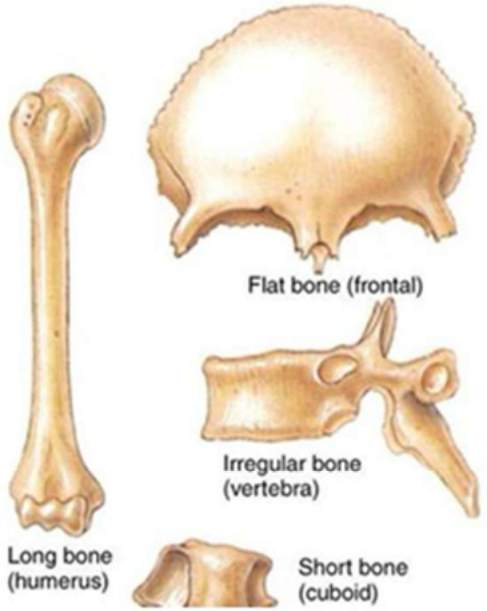
- Irregular: supporting weight, dissipating loads, protecting the spinal cord, contributing to movement, and providing sites for muscular attachment.

  e.g: carpals, ossicles

Anatomical classification of bones

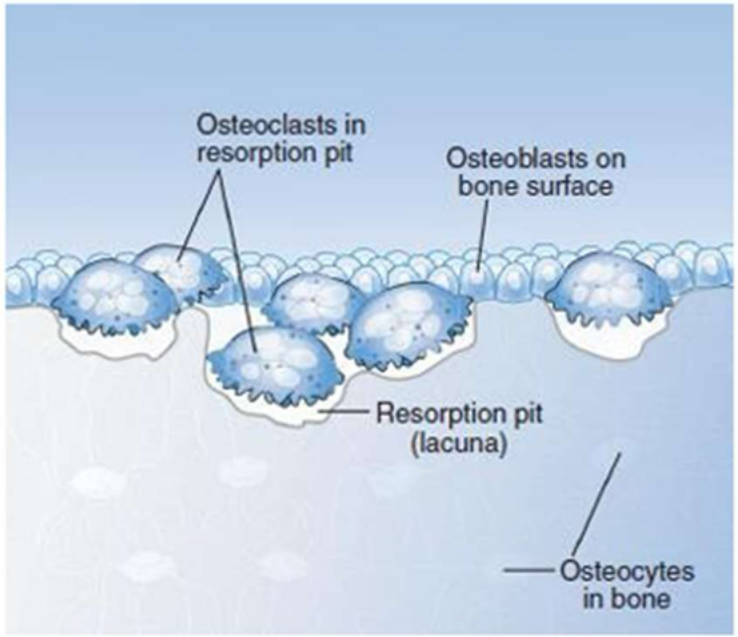






Cells making up bone

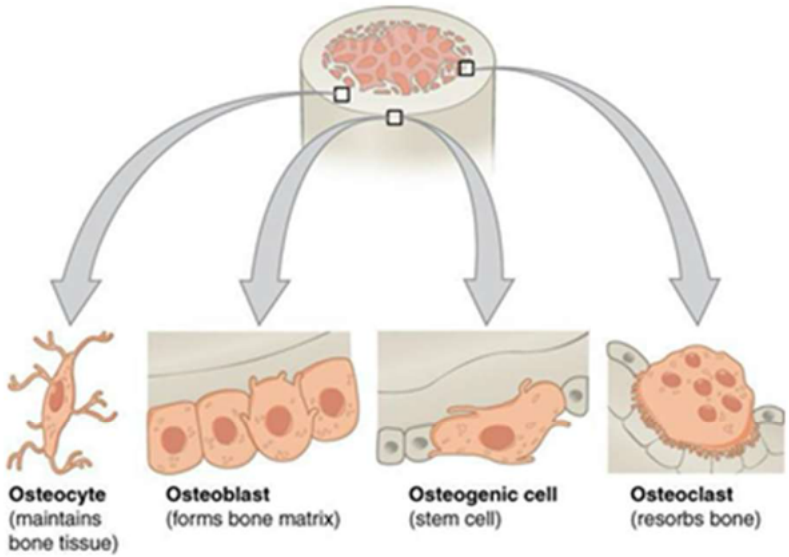
- There are **three** types of cells which make up the bone.
  - Osteoblasts: (bone-forming cells)
  - Osteocytes:(bone-maintaining cells)
  - Osteoclasts: (bone-resorbing cells)



Cells making up bone

Osteoblasts: (bone-forming cells)

- Immature, unspecialized cells which give rise to other bone cells.
- These are the stem cells that will differentiate to become one of the specialized cells.
- They produce a **new bone** called osteoid.
- These cells are also responsible for the calcification of the bone.



Cells making up bone

Osteocytes:(bone-maintaining cells)

- Mature and specialized cells; these cells form the **solid structure** of the bone.
- Helps to maintain bones as living tissue
- Some of the osteoblasts are trapped within the formation of the new bone and develop into osteocytes.

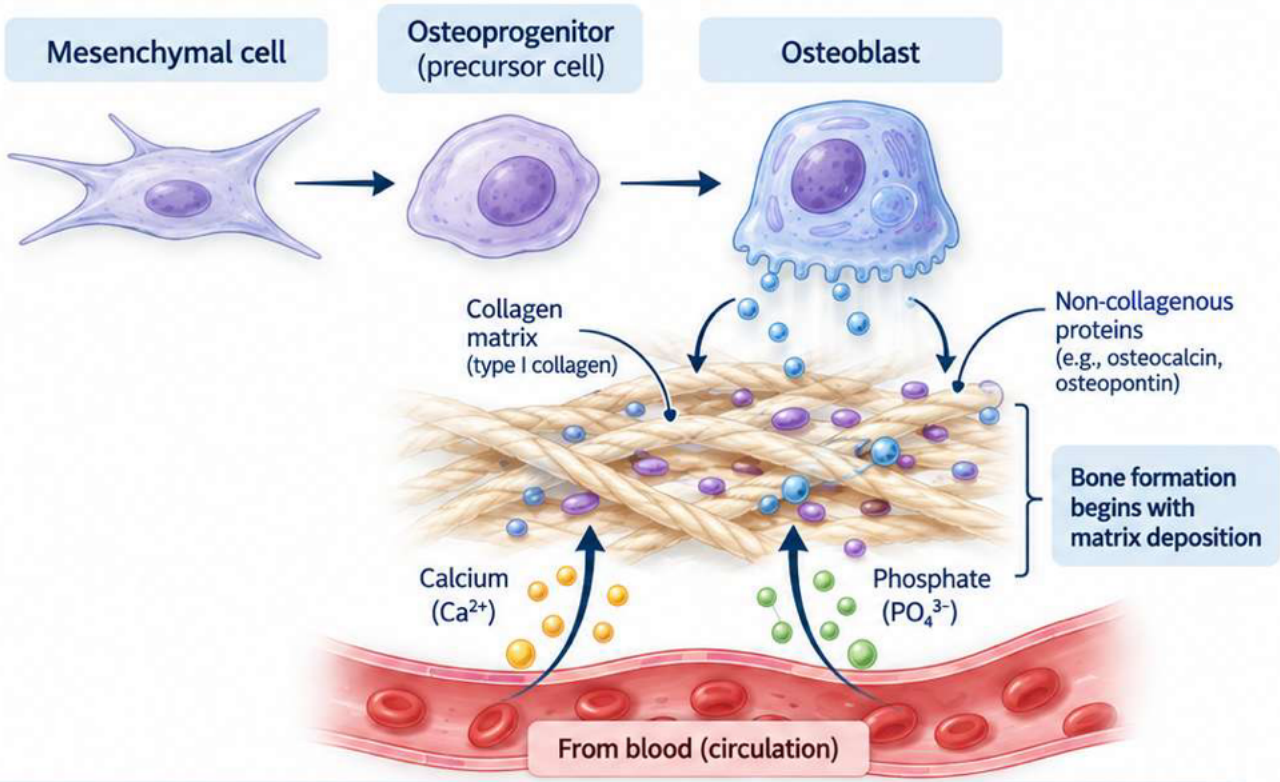
Osteoclasts: (bone-resorbing cells)

- They work to dissolve the bone in areas of micro fracture.
- Scavenger cells which remodel the bone; these cells use enzymes and acid to break down bone tissue, allowing the shape of the bone to be changed to suit the needs of the body, such as during healing or during normal growth.

Bone Cells I: Osteoprogenitors and Osteoblasts

Biomedical Materials — Week 2

- Mesenchymal origin
- Osteoprogenitors = precursor cells
- Differentiate into osteoblasts
- Osteoblasts form new bone
- They secrete collagen matrix



Learning focus: Composition | Cells | Hierarchy

Bone Cells II: Maintenance and Remodeling

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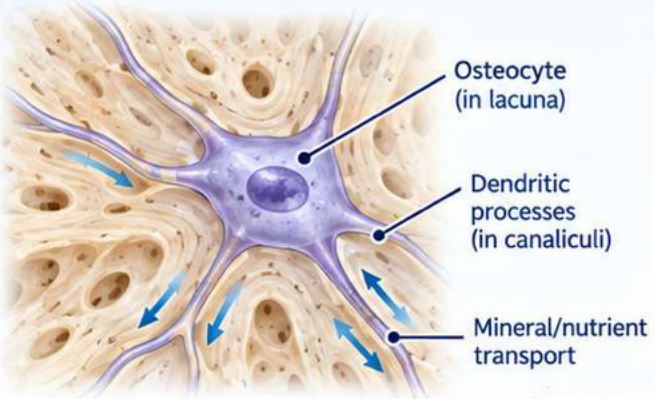
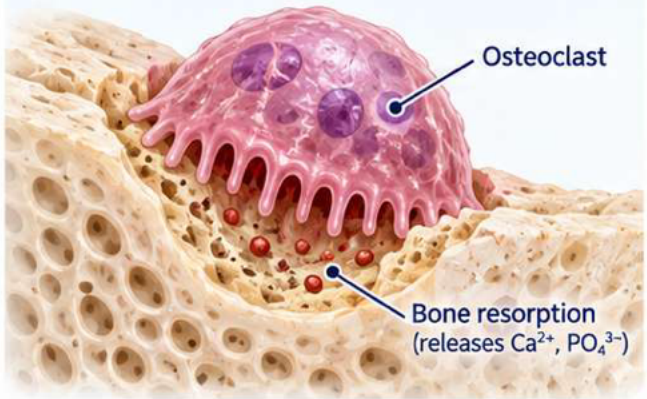
Osteocytes

- Maintain bone
- Mineral transport



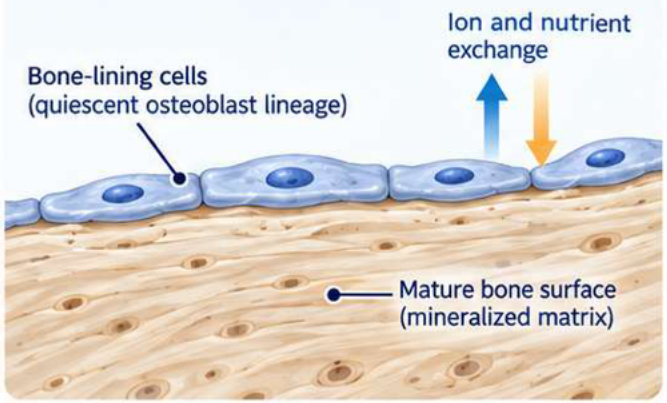
Osteoclasts

- Resorb bone

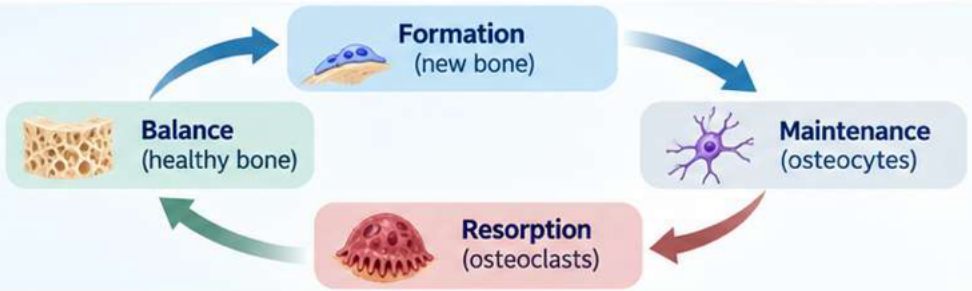


Bone-lining cells

- Regulate exchange



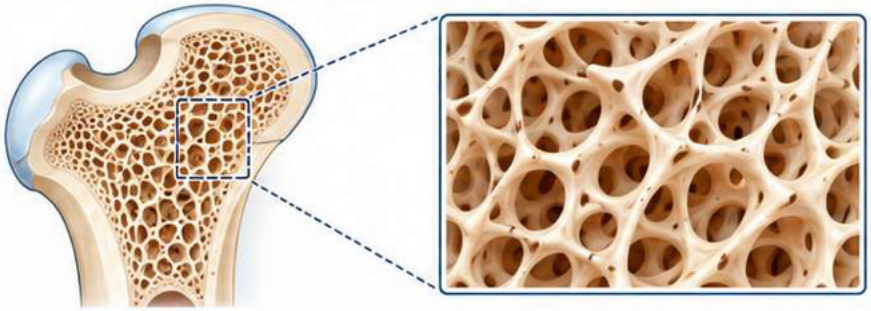
Balanced remodeling = healthy bone



Macro and Micro Structure of Bone

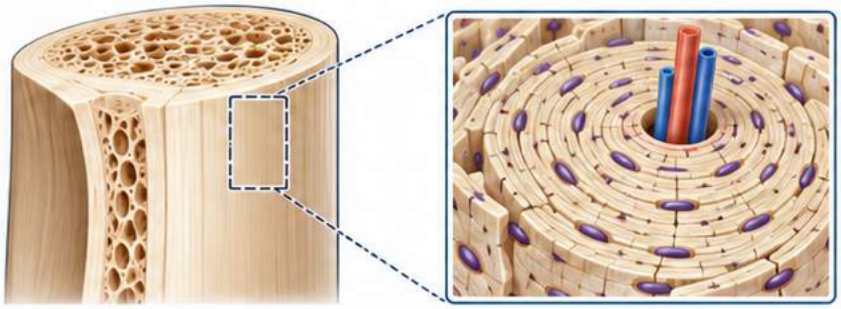
Biomedical Materials — Week 2

Spongy / Trabecular bone



- ~20% of total bone
- High porosity
- More blood vessels
- Trabecular network
- Functions mainly in compression

Compact / Cortical bone

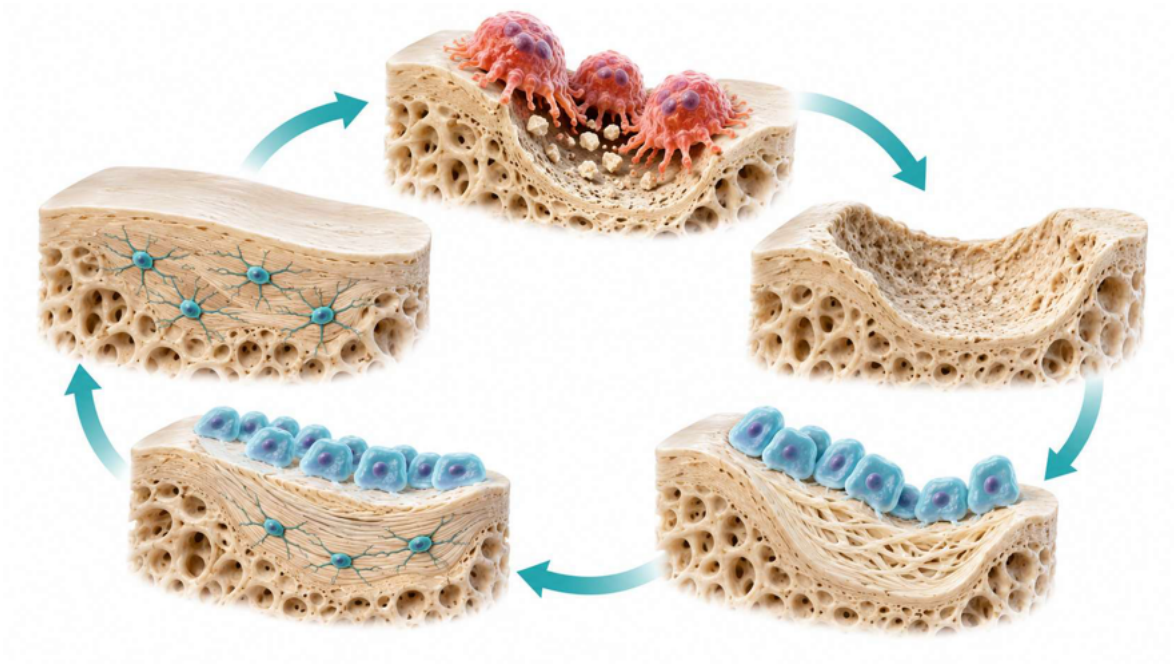


- ~80% of total bone
- Dense and less porous
- Fewer blood vessels
- Osteons as repeating units
- Functions in tension, compression, torsion

Learning focus: Composition | Cells | Hierarchy

Bone Remodeling Cycle

Living bone repairs and adapts through coordinated cell activity



1. Activation
Signals recruit remodeling cells to a local region.

2. Resorption
Osteoclasts remove old or damaged bone.

3. Formation
Osteoblasts deposit new organic matrix.

4. Mineralization
Matrix mineralizes and becomes functional tissue.

Design implication
A scaffold or implant surface should support the intended remodeling response.

Modeling vs. Remodeling

Two similar words, different meanings

Bone modeling

Shape and architecture change mainly during growth and adaptation.

Example: bone geometry adjusts to developmental and mechanical demands.

Bone remodeling

Old or damaged bone is removed and replaced at the same location.

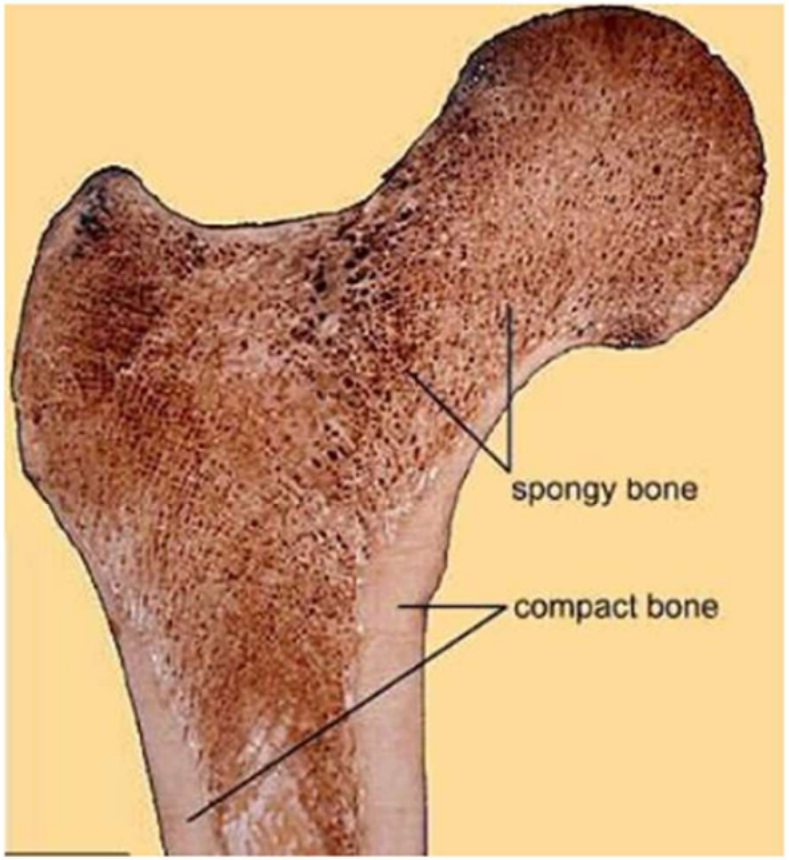
Example: osteoclast resorption followed by osteoblast formation.

For Week 2: understand the biological idea.
For Week 3: we will begin mechanical models and viscoelastic behavior.

Structure of the bone

1. **Compact/cortical Bone (Outer Layer):**

- Compact mineralized connective tissue with **low porosity** that is found in the shafts of long bones.
- 5–30% of bone volume occupied by nonmineralized tissue.
- Dense, Smooth, and Solid to naked eye



Structure of the bone

2. Spongy/cancellous/trabecular Bone (Inner Layer):

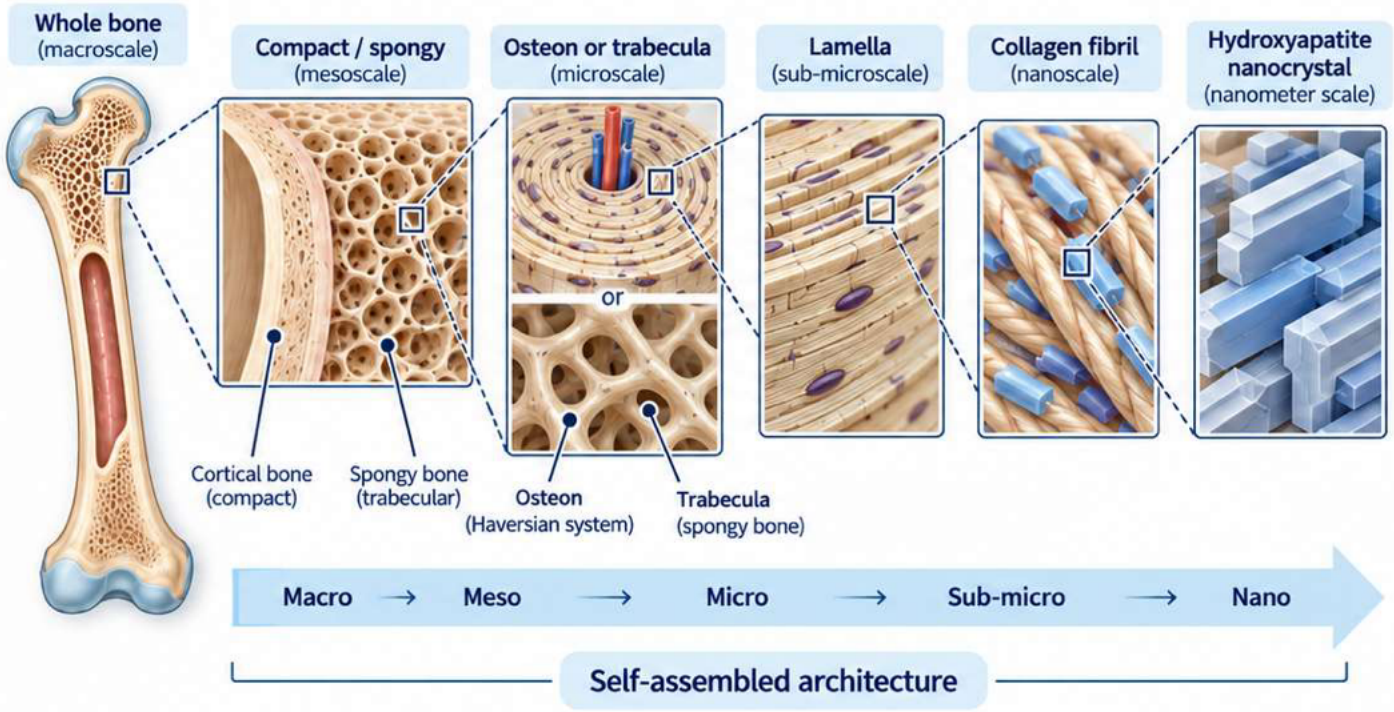
- Less compact mineralized connective tissue with high porosity that is found in the ends of long bones and in the vertebrae.
- >30% of bone volume occupied by nonmineralized tissue.
- Open spaces between trabeculae are filled with red or yellow bone marrow.
- Hole-y (like a honeycomb)

Both cortical and trabecular bone are anisotropic; that is, they exhibit different strength and stiffness in response to forces applied from different directions

Hierarchical Tactics of Bone Tissue

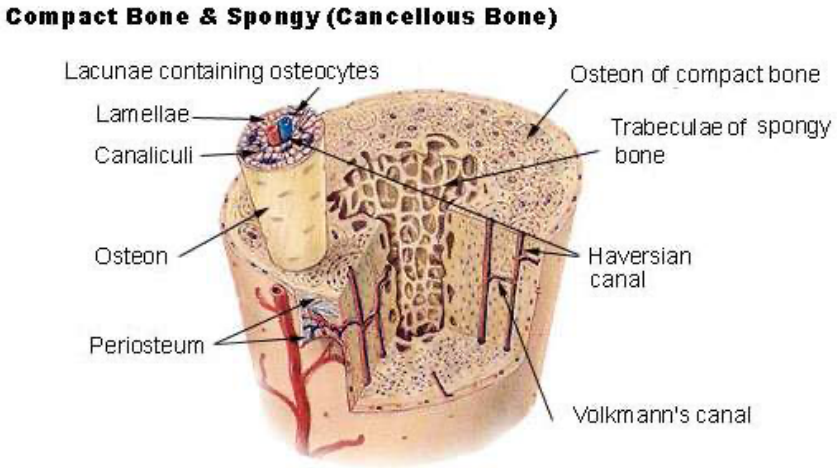
Biomedical Materials — Week 2

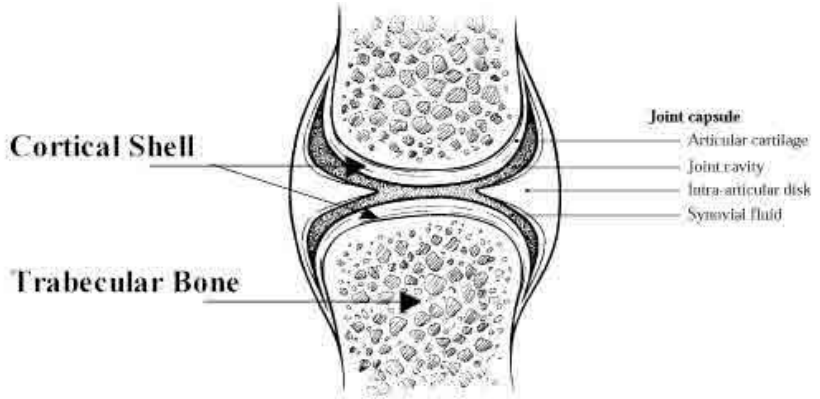
- Multi-scale organization
- Macro to nano assembly
- Structure drives function
- Remodeling improves alignment
- Hierarchy gives strength and toughness



Learning focus: Composition | Cells | Hierarchy

Cortical and Trabecular Bone



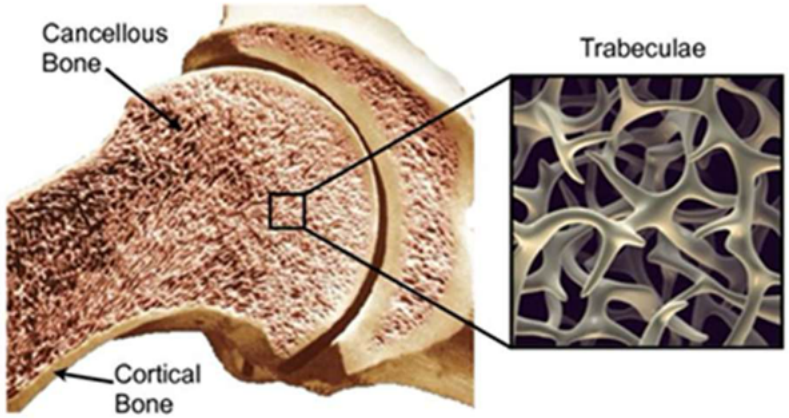


Bone Density

- The amount of bone minerals in bone tissue.
- The **density** of different bones and their **mechanical properties** of bone vary from type to type.

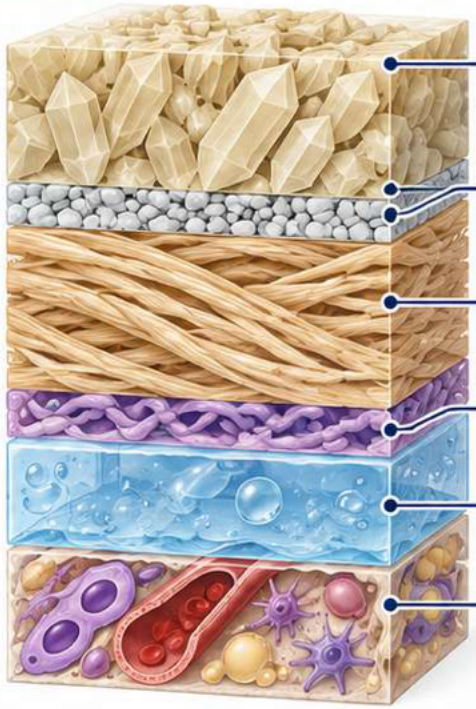
- **Cortical bone:** has a density of approximately 2g/cm3, with 5 –30% porosity.

- **Spongy bone:** has a density of approximately 0.07 –1g/cm3, with porosity > 30%.



Components of Bone Tissue — Overview

Biomedical Materials — Week 2



- Hydroxyapatite ~60 wt%
- Carbonate ~4 wt%
- Collagen ~20 wt%
- Non-collagenous proteins ~3 wt%
- Water ~9 wt%
- Cells, lipids, vessels, trace components ~4 wt%

- Multi-component connective tissue
- Minerals + organic matrix
- Water supports transport
- Cells maintain and remodel
- Organized from nano to macro

Mineral phase
(hydroxyapatite + carbonate)

Organic phase
(collagen + non-collagenous proteins)

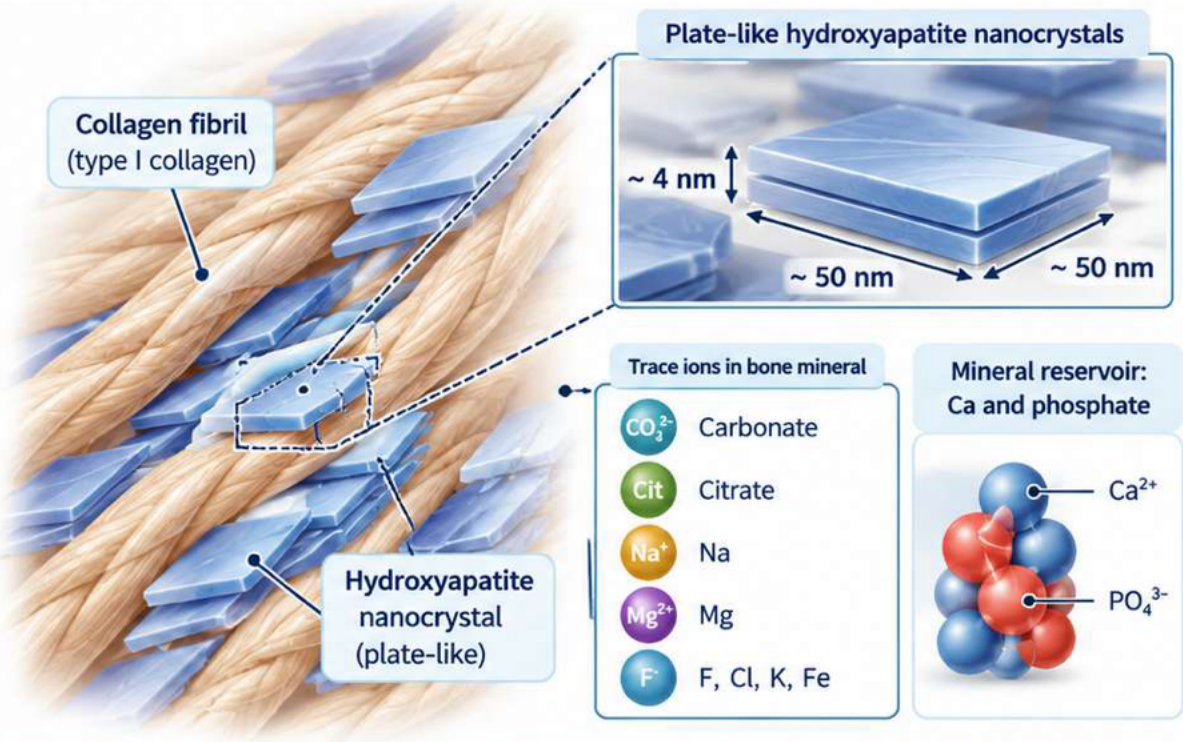
Water
(structural and transport)

Cellular part
(cells, lipids, vessels, trace components)

Inorganic Phase: Bone Mineral

Biomedical Materials — Week 2

- Mainly nanophase hydroxyapatite
- Plate-like nanocrystals
- Approx. 4 × 50 × 50 nm
- Trace ions enrich the mineral
- Provides rigidity and stiffness

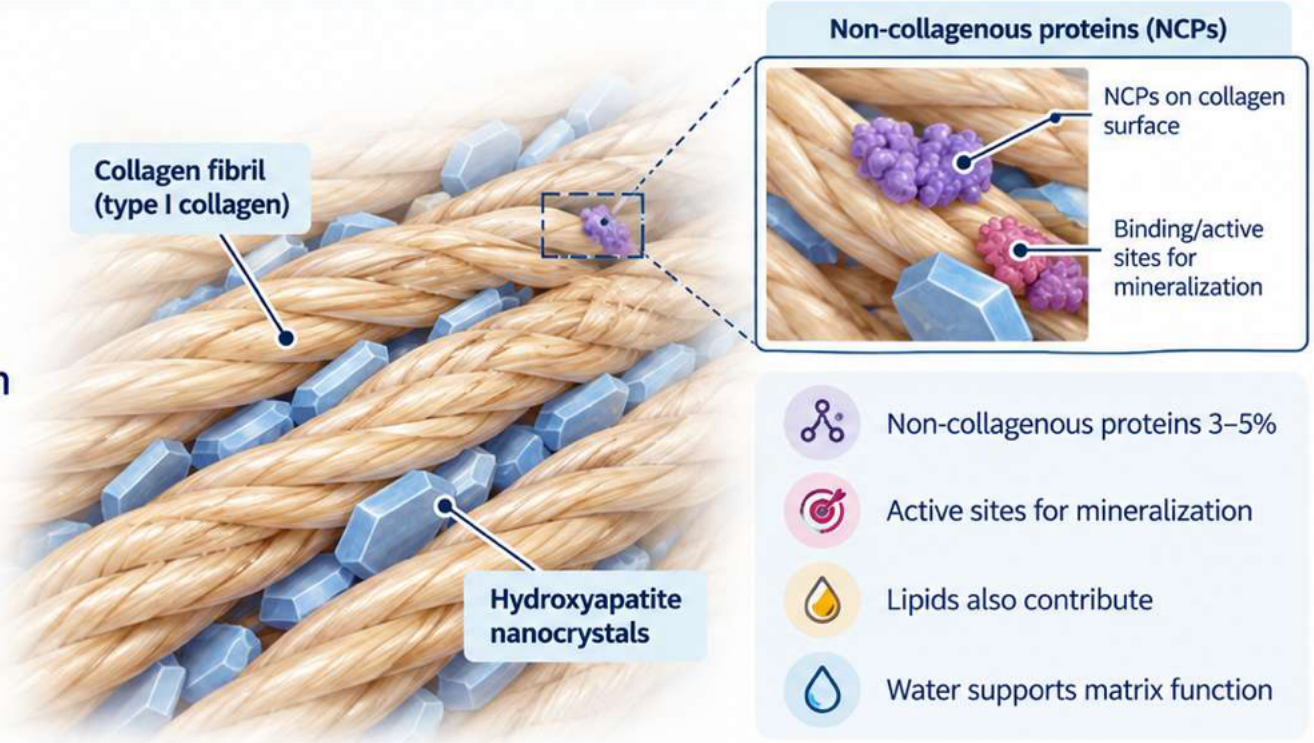


Learning focus: Composition | Cells | Hierarchy

Organic Phase: Collagen and Matrix

Biomedical Materials — Week 2

- Collagen is the main protein
- Type I collagen framework
- Fiber diameter 100–2000 nm
- Gives tensile strength
- Adds flexibility to bone



- Non-collagenous proteins 3–5%
- Active sites for mineralization
- Lipids also contribute
- Water supports matrix function

Learning focus: Composition | Cells | Hierarchy

Bone as a Natural Composite

Collagen and mineral phases work together

Collagen
Organic fibers contribute ductility and toughness.

Mineral apatite
Inorganic crystals contribute stiffness and compressive strength.

Interfaces
Load transfer occurs through nanoscale bonding and interlocking.

Hierarchy
Composite behavior changes with scale: nano → micro → tissue → whole bone.

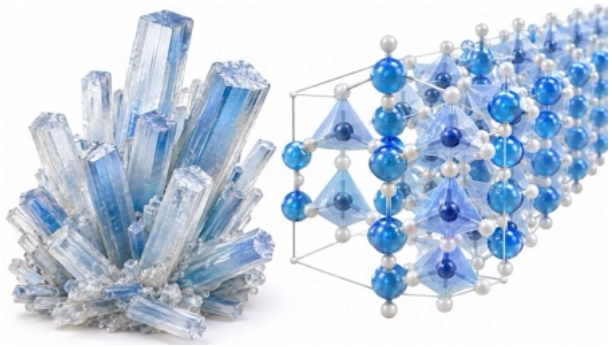
Chemical Composition of Bone Mineral

Biological apatite is complex, substituted, and imperfect

Ideal hydroxyapatite
$Ca_{10}(PO_4)_6(OH)_2$

Bone mineral
Poorly crystalline apatite containing substitutions and defects.

Common ionic groups
Ca^{2+}, PO_4^{3-}, CO_3^{2-} plus smaller quantities of Mg^{2+}, Na^+, K^+, Cl^-, F^-.

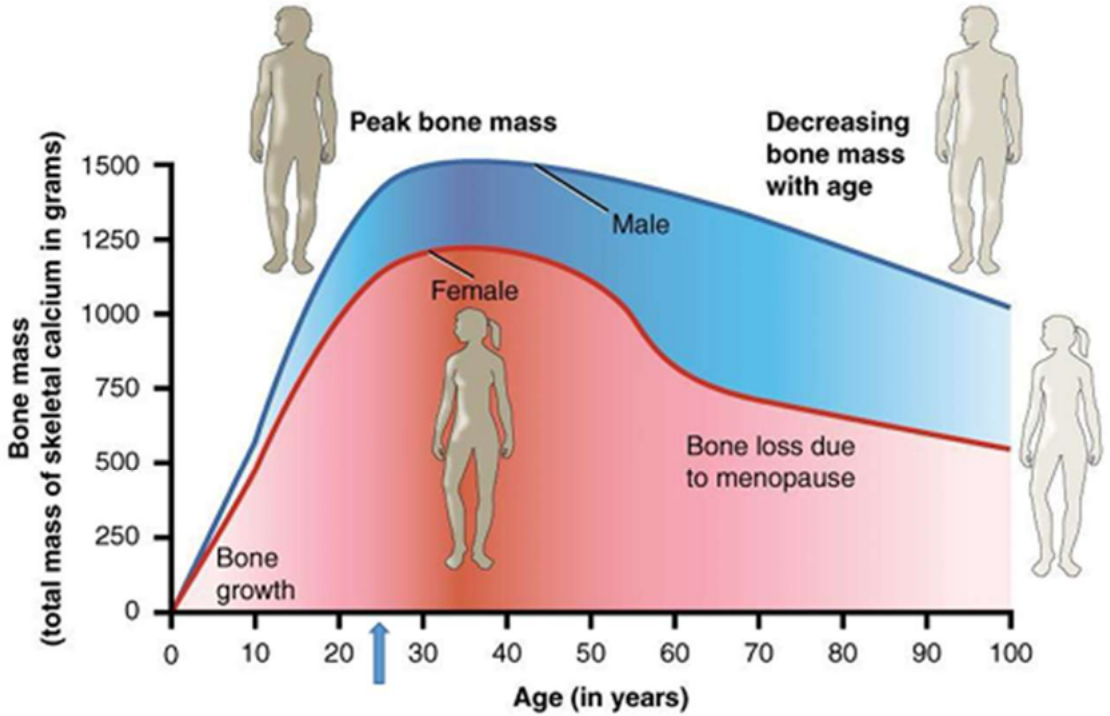


Why this matters: crystal chemistry affects solubility, remodeling, coating behavior, and biological response.

Bone Density

Factors affecting bone density:

- Hormonal diseases (mainly those affecting oestrogen, testosterone and cortisol)
- Local injuries/inflammation
- Lack of physical activity
- Malnutrition (lack of calcium)
- Genetics: (white vs black)
- Lack of vitamin D.
- Ageing



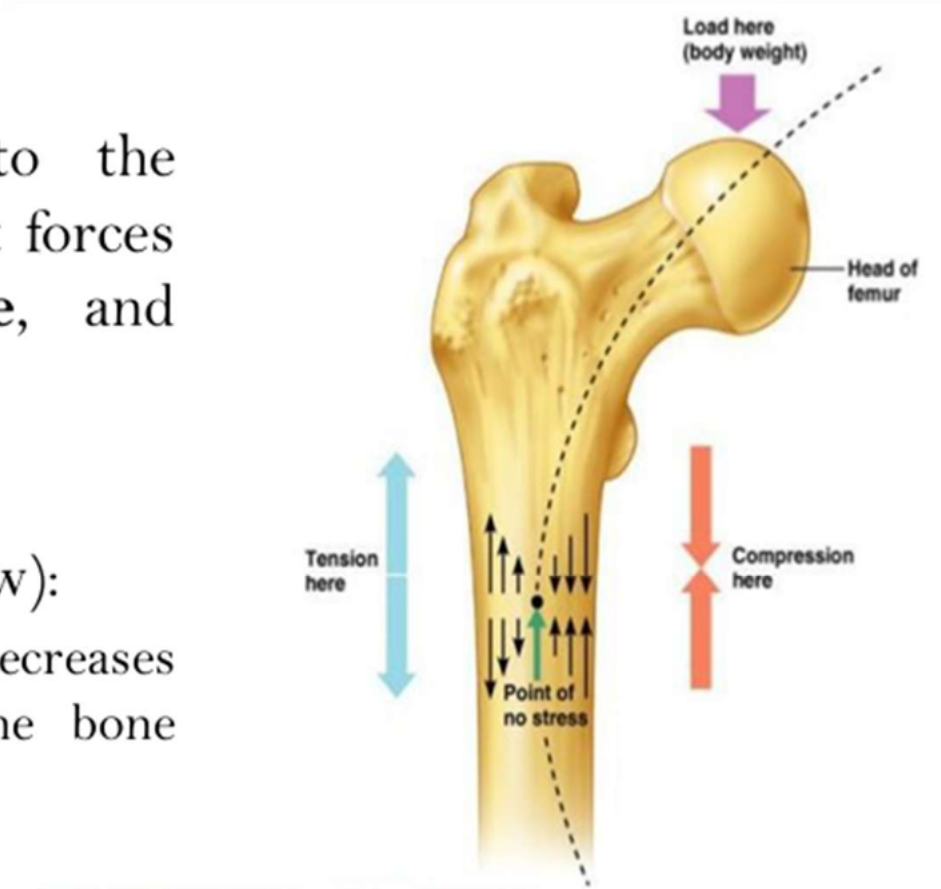
Mechanical properties of bone

- The mechanical properties of bone are as **complex** and varied as its **composition**.
- The mechanical properties also vary with **age** and **gender** and with the **location of the bone** (humerus versus the tibia).
- Measurement of bone **strength**, **stiffness**, and **energy** depends on both the **material composition** and the **structural properties** of bone.

Bone response to stress

- Bone responds dynamically to the presence or absence of different forces with changes in **size**, **shape**, and **density**.

- Julius Wolff in 1892 (Wolff's Law):
  - The bone strength increases and decreases as the **functional forces** on the bone increase and decrease.



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Bone response to stress

Bone Modeling and Remodeling:

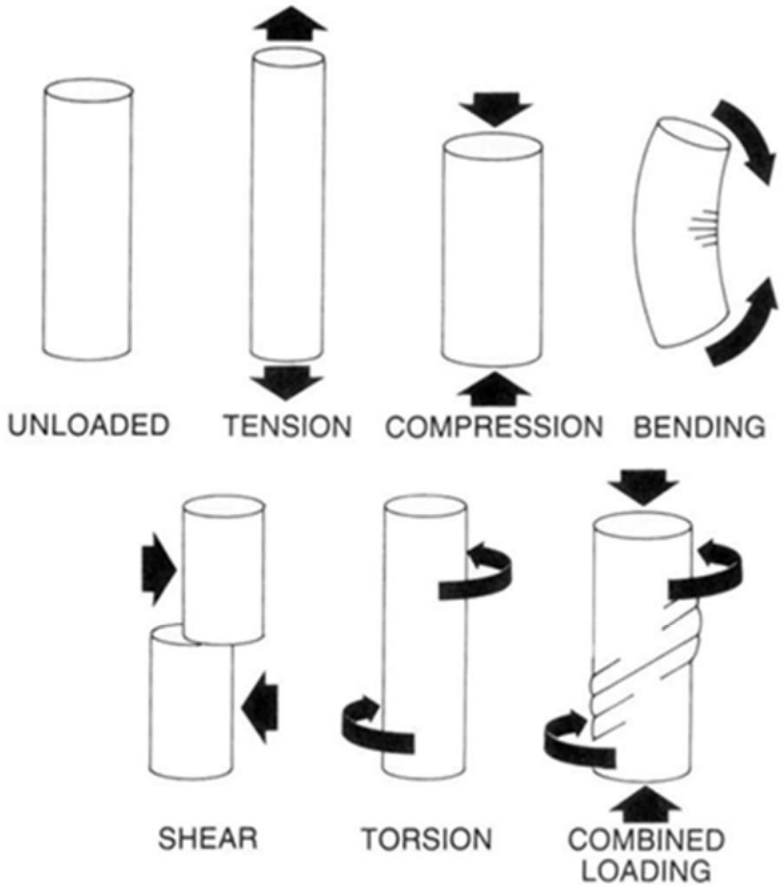
- The **densities**, the **shapes** and **sizes** (to a much lesser extent) of the bones of a given human being are a function of the **magnitude** and **direction** of the **mechanical stresses** that act on the bones.
- Dynamic mechanical loading causes bones to deform or strain, with larger loads producing higher levels of strain.
- These strains are translated into changes in bone shape and strength through a process known as **remodeling.**

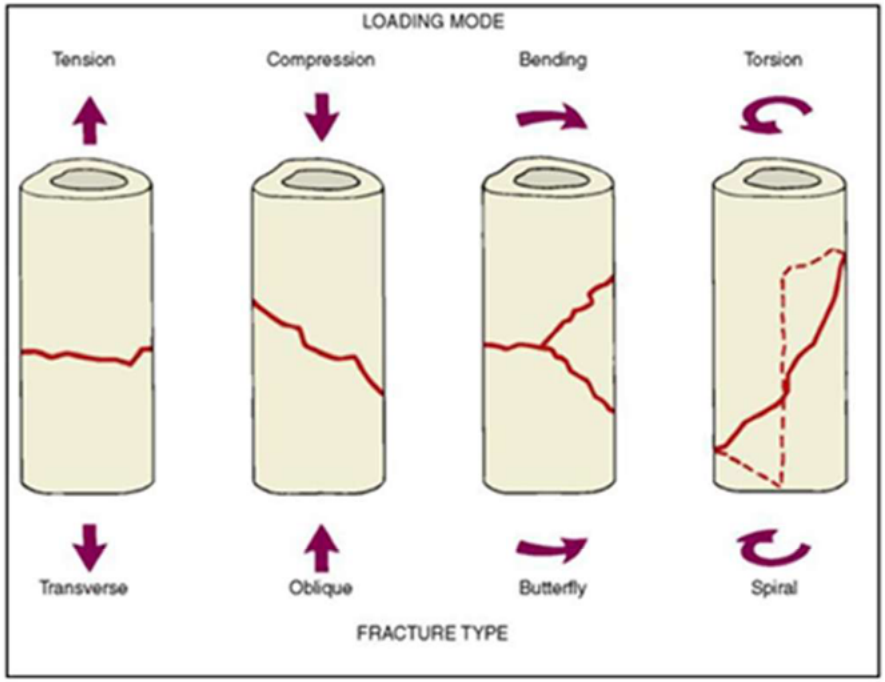
*** Bone remodeling Vs Bone modeling**

Load applied to Bone

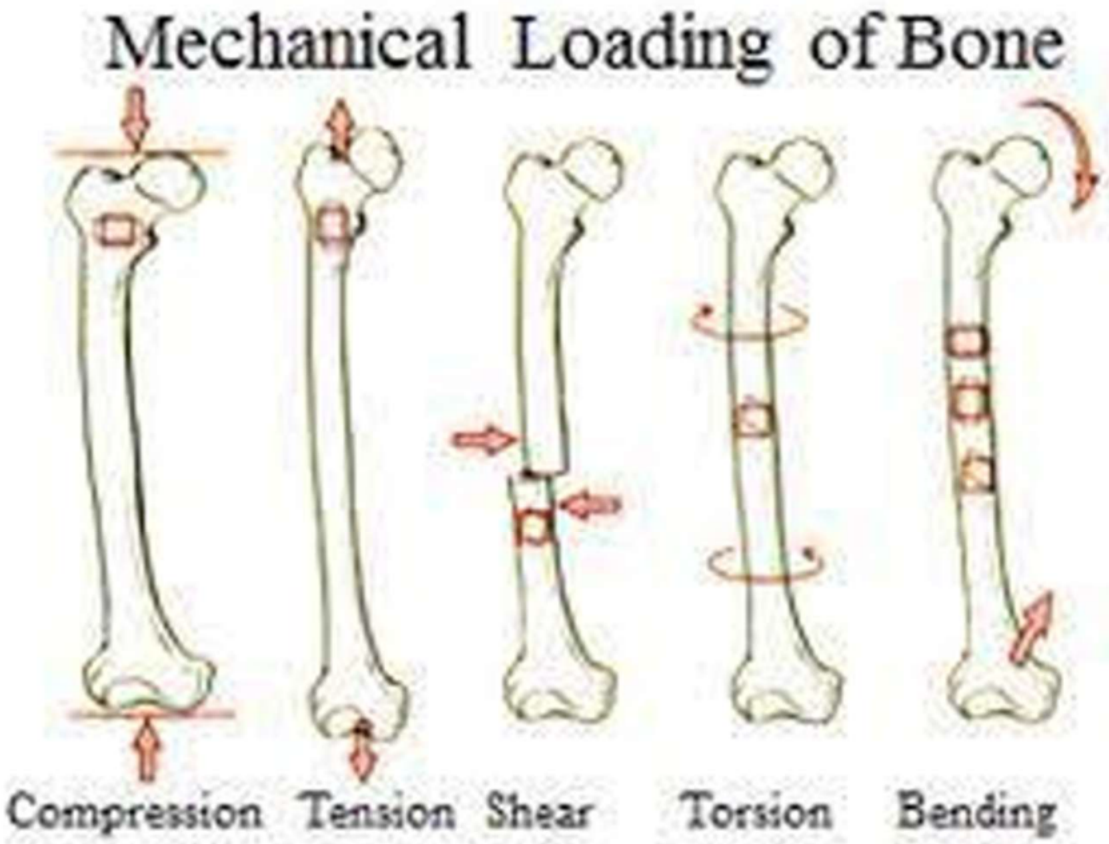
- The skeletal system is subject to a variety of applied forces as bone is loaded in various directions.
- Loads are produced by **weight bearing**, **gravity**, **muscular forces**, and **external forces**.
- Six modes of loading
- During activity usually a combination of loading modes takes place.
- Each mode of loading produces a distinctive **fracture pattern** if the ultimate strength of the bone is exceeded.

Load applied to Bone



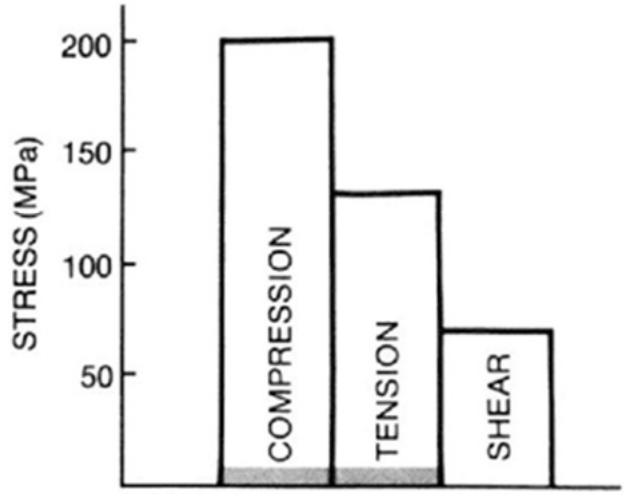


Load applied to Boneloading



Load applied to Bone

- Bone is ***anisotropic***: its strength varies according to the orientation of the loading.
- In general, bone is **stronger** in **compression** than in **tension**, and **stronger** in **tension** than in **shear**.



Comparative Loading Strengths

Load applied to Bone

Different Types of Loads Acting on Bone

Load	Type of Force	Source	Stress/Strain
Compression	Presses ends of bones together to cause widening and shortening	Muscles, weight bearing, gravity or external forces	Maximal stress on the plane perpendicular to the applied load
Tension	Pulls or stretches the bone to cause lengthening and narrowing	Usually pull of contracting muscle tendon	Maximal stress on the plane perpendicular to the applied load
Shear	Force applied parallel to surface, causing internal deformation in an angular direction	Compressive or tension force application or external force	Maximum stress on the plane parallel to the applied load
Bending	Force applied to the bone having no direct support from the structure	Weight bearing or multiple forces applied at different points on the bone	Maximum tensile forces on the convex surface of the bent member and maximum compression forces on the concave side
Torsion	Twisting force	Force applied with one end of the bone fixed	Maximum shear stress on both the perpendicular and parallel to axes of bone with tension and compression forces also present at an angle across the surface

Strength of Bones

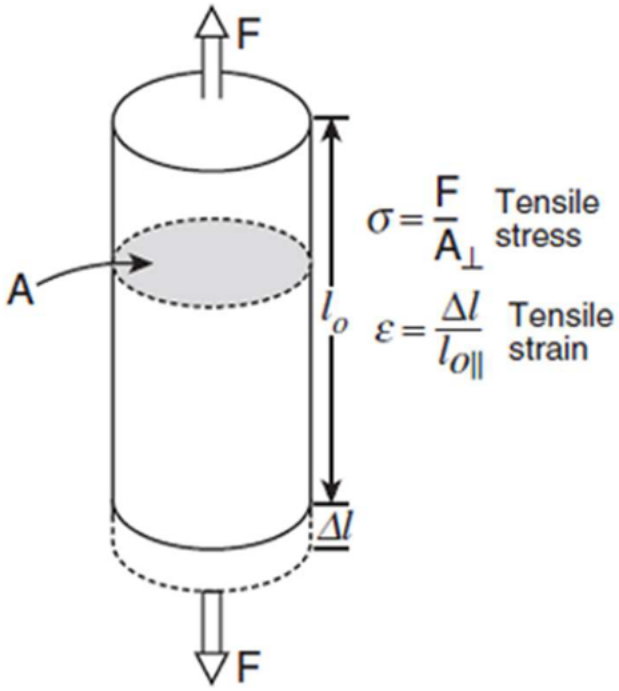
Stress Strain behaviour:

- The ***stress*** is defined as measure of the force distribution over a given cross-sectional area:

$$\boldsymbol{stres\ (\sigma) = \frac{Force\ (F)}{Area\ (A)} \quad \left(\frac{N}{m^2}\right)}$$

- The relative deformation of an object in response to an applied load is called ***strain***:

$$\boldsymbol{Strain\ (\varepsilon) = \frac{change\ in\ length}{original\ length} = \frac{\Delta \iota}{\iota}}$$

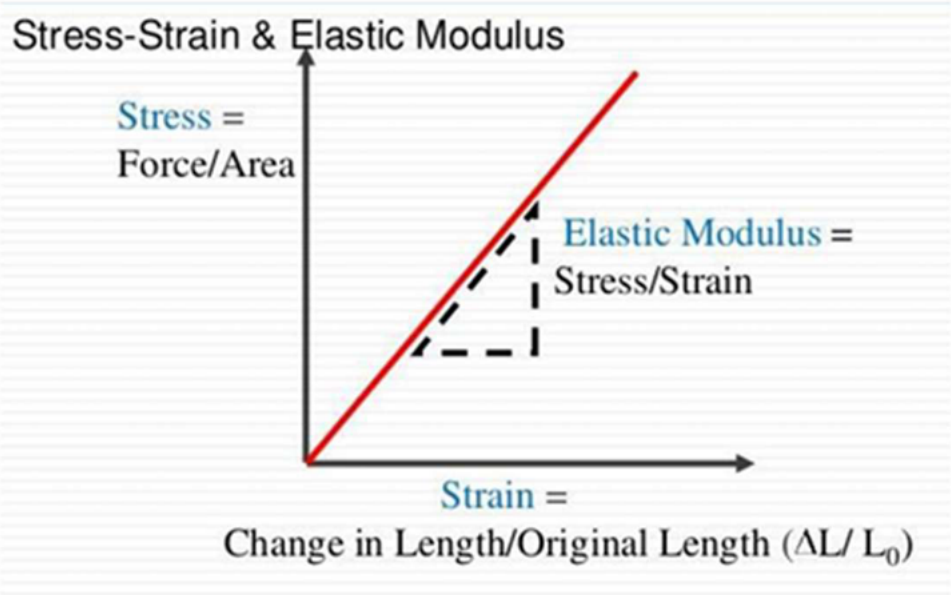


Strength of Bones

- Youngu's Modulus

$$\frac{Stress}{Strain} = \frac{\frac{F}{A}}{\frac{\Delta L}{L_0}} = E$$

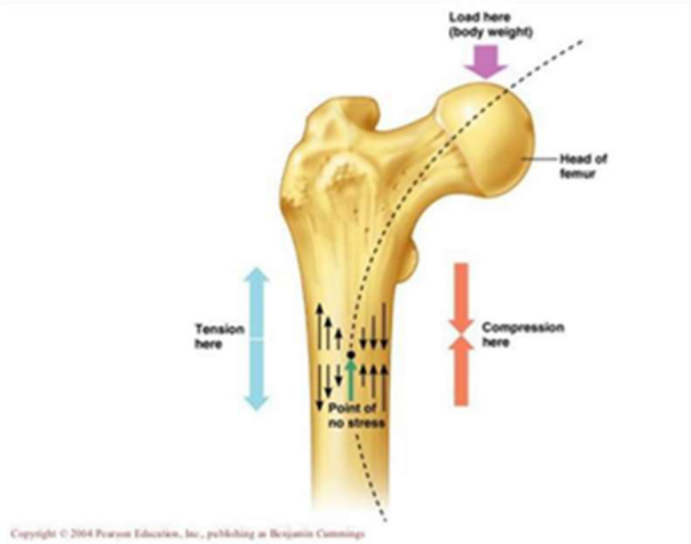
.

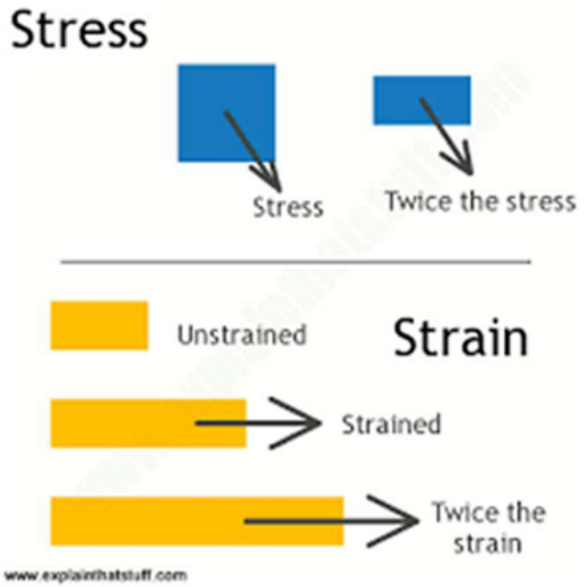


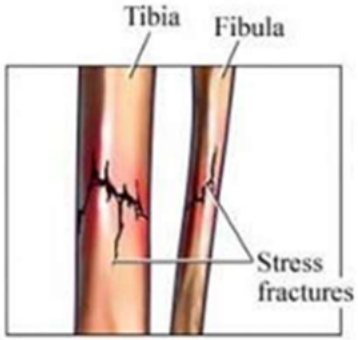
Strength of Bones

- The relative deformations created at any point are referred to as the strains at that point.

$$Strain = \frac{change\ in\ length}{original\ length} = \varepsilon = \frac{\Delta \iota}{\iota}$$

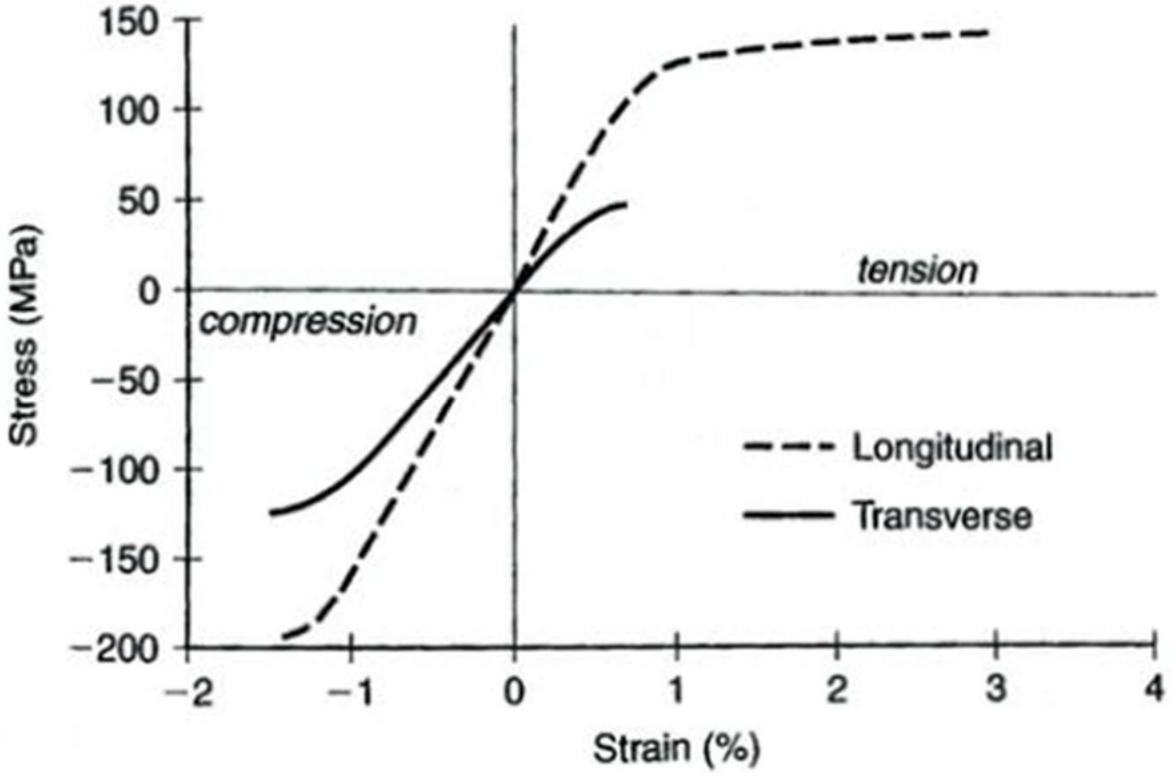




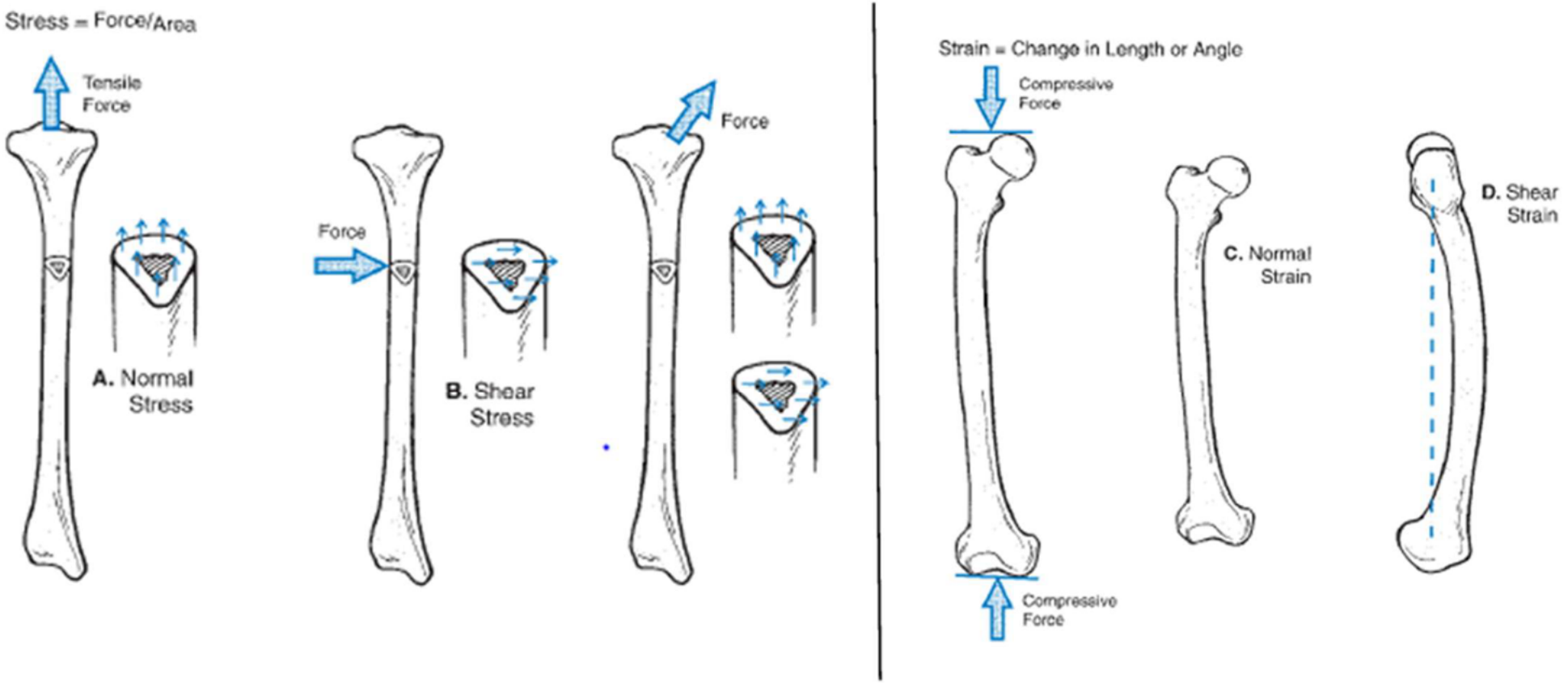


Strength of bones

Stress-strain curves of typical cortical bone



Strength of bones

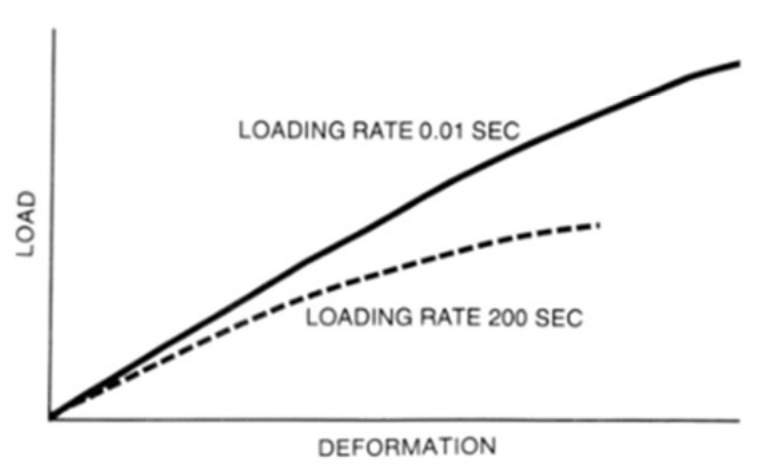


Strength of Bones

- Elastic Modulus (GPa) of Common Materials in Orthopaedics
  - Stainless Steel 200
  - Titanium 100
  - Cortical Bone 7-21
  - Bone Cement 2.5-3.5
  - Cancellous Bone 0.7-4.9

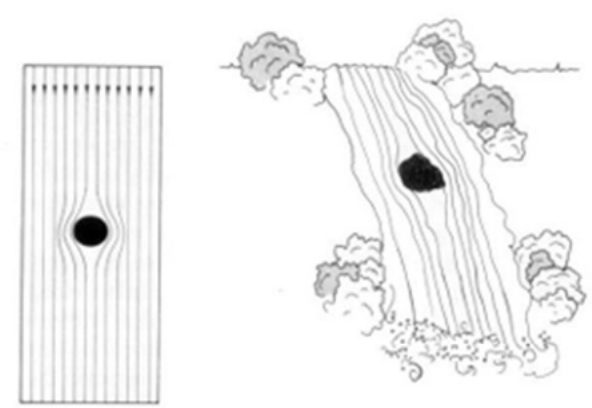
Strength of bones

- Bone is *visco-elastic*, meaning the mechanical properties such as stiffness change as the strain rate changes.
- Stiffness increases for higher strain rates.
- This improves the bone's ability to withstand high-impact loading, such as might occur during running.



Stress raisers in bone

- In engineering terms stress is considered to "flow" in a material.
- Any sudden changes in the material result in a change in the flow pattern.



- **Fractures** are more likely to occur at stress concentrated sites if the bone is loaded.

Stress risers in bone

Particular examples of stress raisers in bone are:

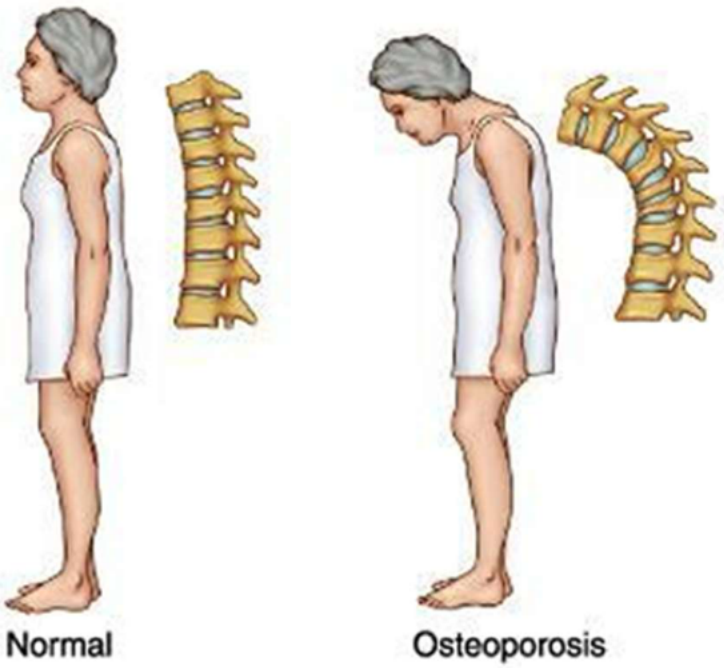
- Holes drilled into bone to secure metal bone plates.
- Pins and screws placed in the bone during surgery.
- Stress fractures and hairline cracks.
- The junction between implants and the bone.

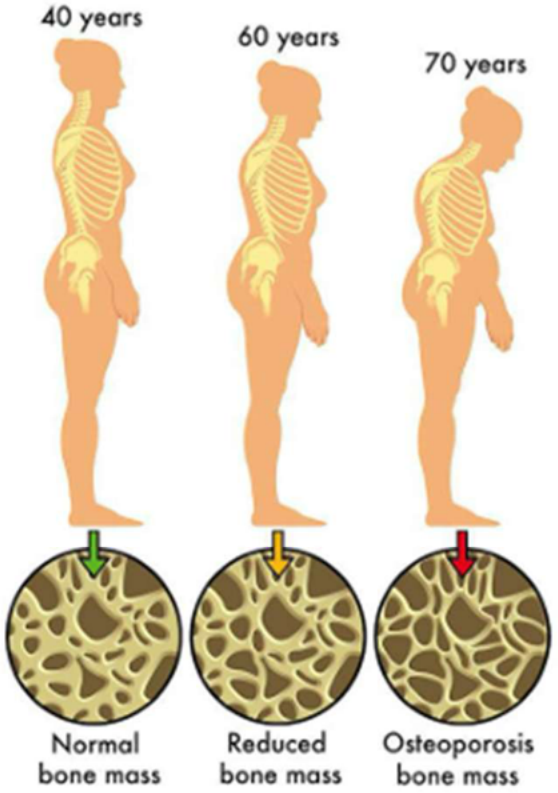
Degenerative processes in bone

***Osteoporosis*:**

- This is a loss of bone density, thereby increasing the bone porosity.
- It is usually hormonally linked, i.e. related to a loss in oestrogen or testosterone levels.

Degenerative processes in bone





Degenerative processes in bone

Bone resorption:

- This is the excessive resorption of bone when the bone is insufficiently loaded.
- It may be caused by Orthopedic implants such as artificial joints, plates and screws, or by a sedentary lifestyle, or physical inactivity.

Common bone injuries

Fractures:

- A fracture is a disruption in the continuity of a bone.

Types of Bone fracture:

Common bone injuries

A- ***Greenstick fracture*** is incomplete, and the break occurs on the convex surface of the bend in the bone.

B- ***Fissured fracture*** involves an incomplete longitudinal break.

C- ***Comminuted fracture*** is complete and fragments the bone.

D- ***Transverse fracture*** is complete, and the break occurs at a right angle to the axis of the bone.

E- ***An oblique fracture*** occurs at an angle other than a right angle to the axis of the bone.

F- ***Spiral fracture*** is caused by twisting a bone excessively.

Fractures of the femur

The mode of loading, as well as the magnitude of the forces applied to the bone, will influence the type of fractures that can be expected

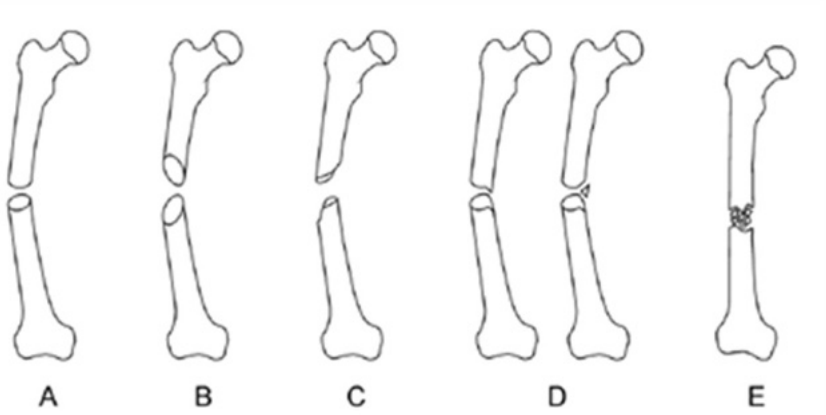
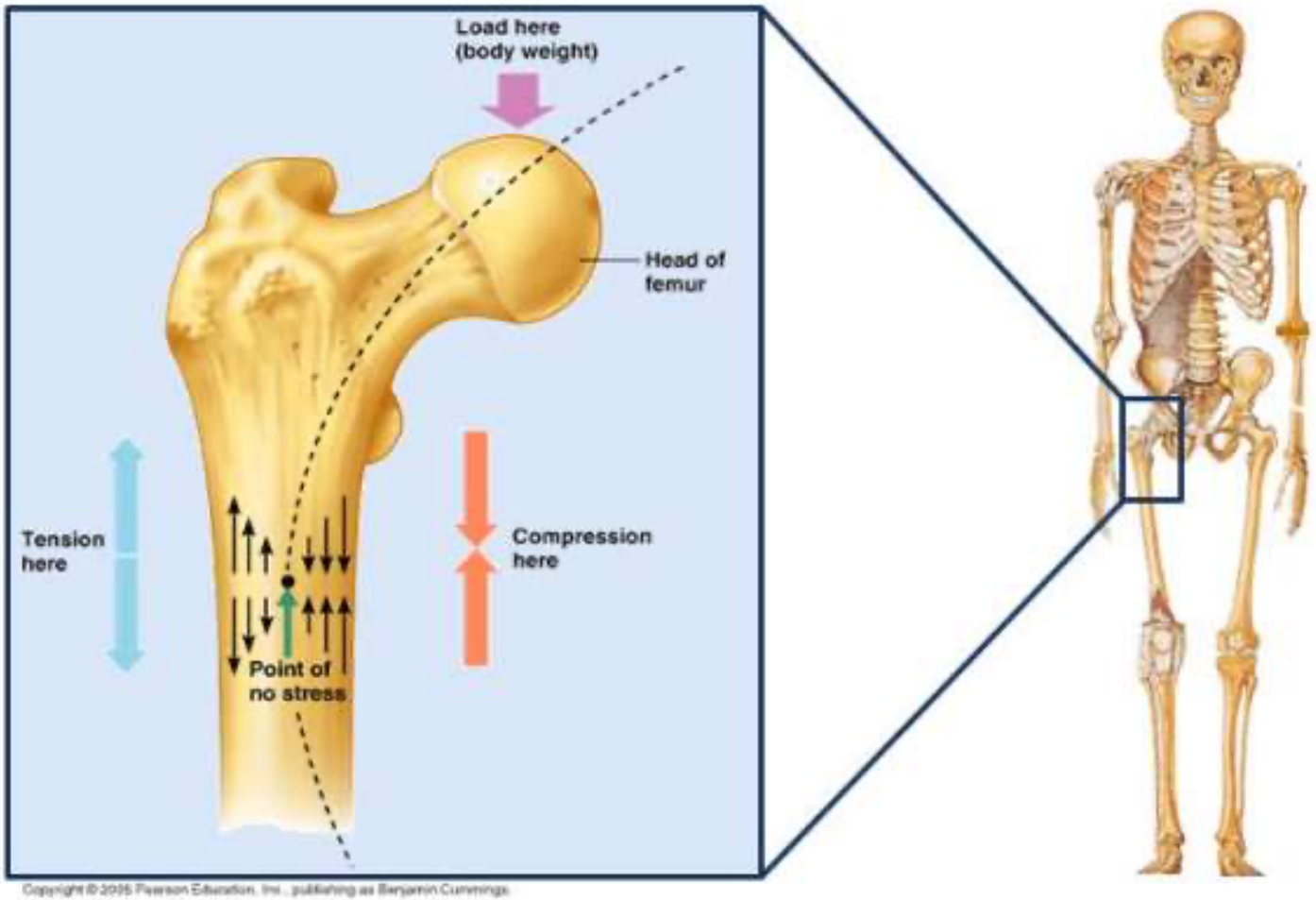


Image	Pattern	Mode	Energy
A	Transverse	Bending	Low
B	Oblique	Compression + Bending + Torsion	Moderate
C	Spiral	Torsion	Low
D	Oblique-transverse or Butterfly	Compression + Bending	Moderate
E	Comminuted	Variable	High



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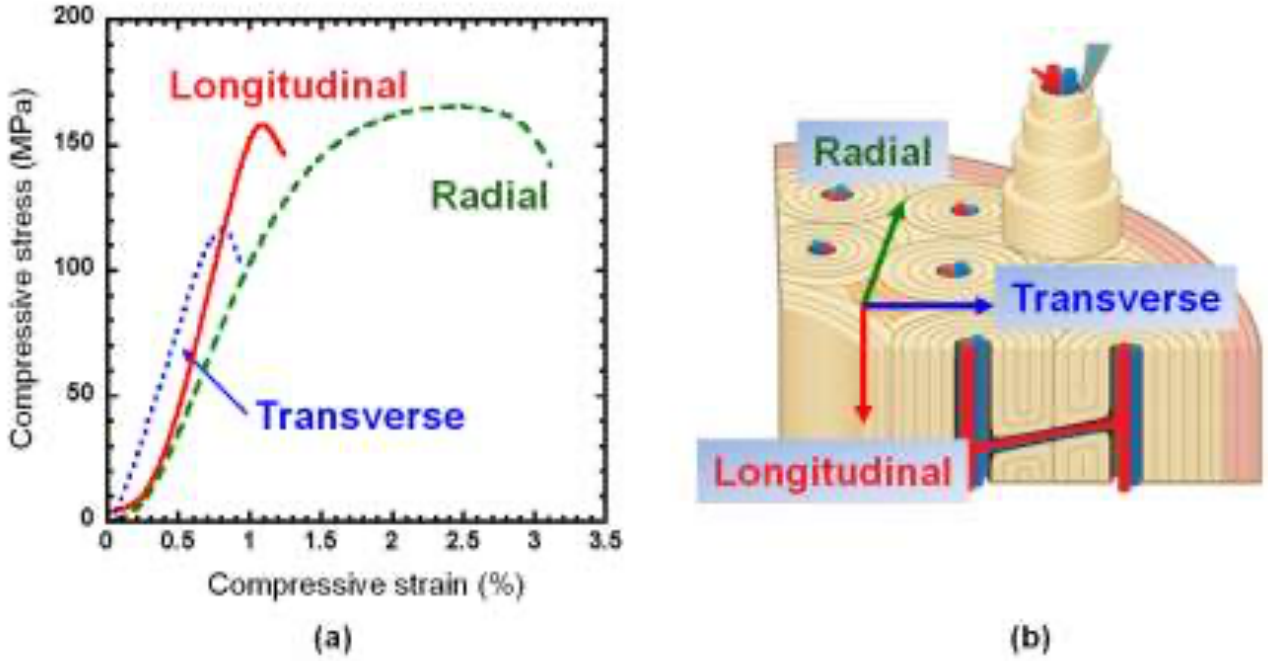
Bone Compressive Strength

Material	Compressive Strength (MPa)
Femur (cortical)	131-224
Tibia (cortical)	106-200
Wood (oak)	40-80
Steel	370

From: Biomechanics of the Musculo-skeletal System, Nigg and Herzog



(a) Compressive stress-strain curves for bovine cortical bone for three anatomical directions. (b) The longitudinal direction is clearly the stiffest bone direction, while the radial one has the maximum toughness. Modified from Novitskaya et al. (2011).

Bone Elastic Modulus. E

Cortical Bone

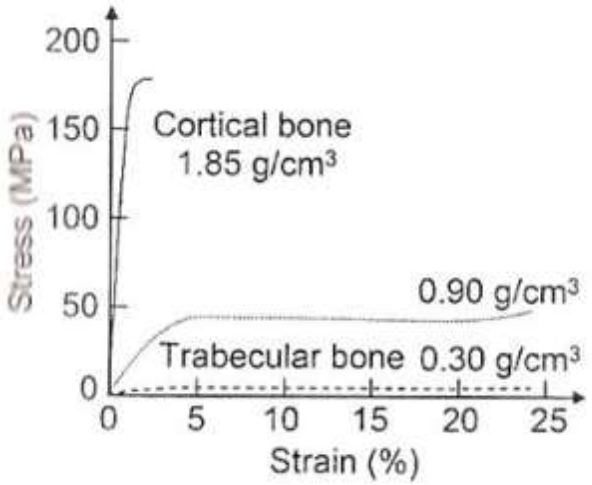
- $E \approx 17.9$ GPa
- $\sigma_{ult} \approx 170$ MPa (compression)
- $\sigma_{ult} \approx 120$ MPa (tension)

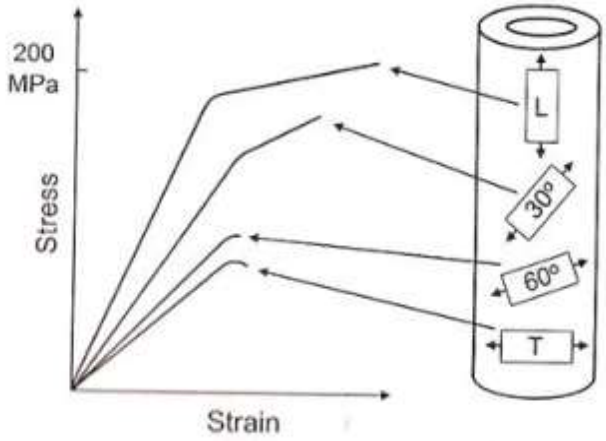
Trabecular Bone

- $E \approx 0.076$ GPa
- $\sigma_{ult} \approx 2.2$ MPa (compression)

Note: Bone is non-homogeneous and assumed orthotropic

(spatial & direction dependence)





Heterogeneity

Variation in microstructural parameter

- porosity
- percentage of mineralization

Modulus and Ultimate strength decreases by half when porosity increases from 5 to 30%

Small increase in percentage mineralization, large increases in both modulus and strength

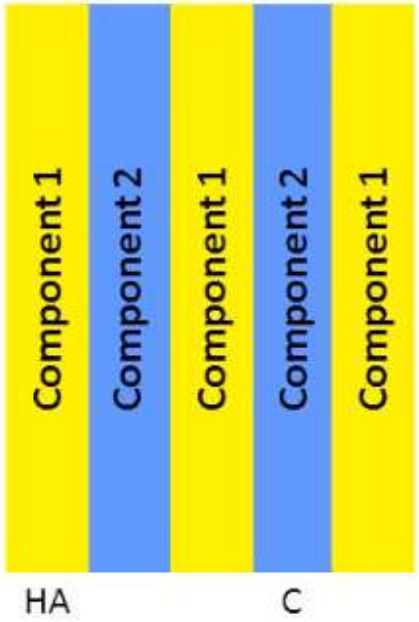
Aging

- Tensile U.stress decreases at a rate of app. 2% per decade
- T.U.Strain decreases by about 10% of its young-value/ decade from 5 %
- Energy to fracture, less for old bone than for younger bone
- Fracture toughness decreases with aging, Old C.B are more brittle than young
- Mineralization does not change much with aging. It is due to collagen changes
- Age related changes – Porosity increases with age
- Strength & ductility decreases with age – Elastic Modulus changes with age
- Fracture mechanics studies- shows decrease in frature toughness with age

Mixture Theory

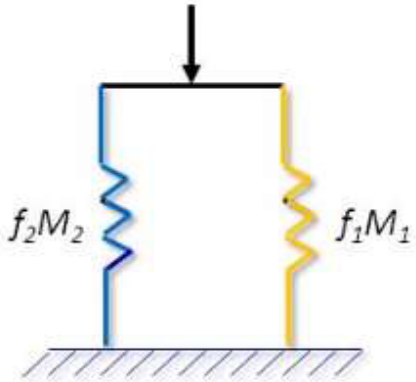
M_i = *Modulus of* i^{th} *constituent*

f_i = *volume fraction of* i^{th} *constituent*



Voigt Upper Bound:

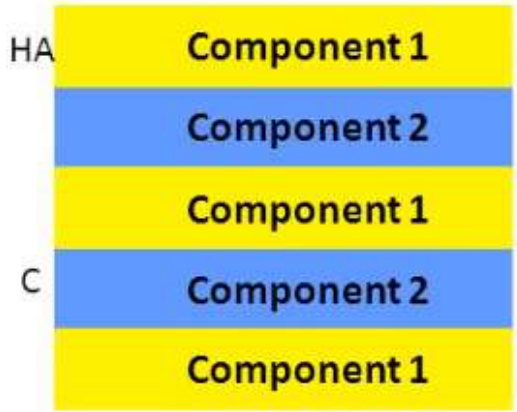
Model as parallel components in mixture



$$M_v = \sum_{i=1}^{N} f_i M_i$$

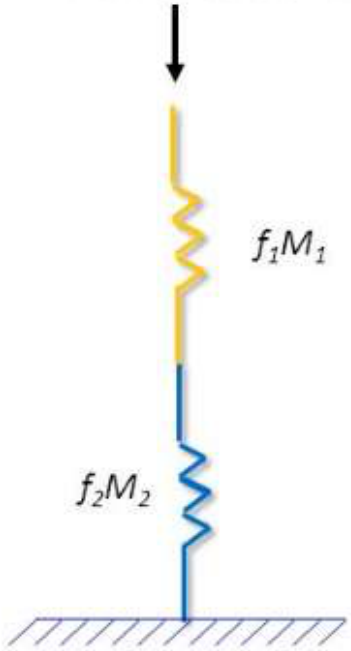
M_i = *Modulus of* i^{th} *constituent*

f_i = *volume fraction of* i^{th} *constituent*



Reuss Lower Bound:

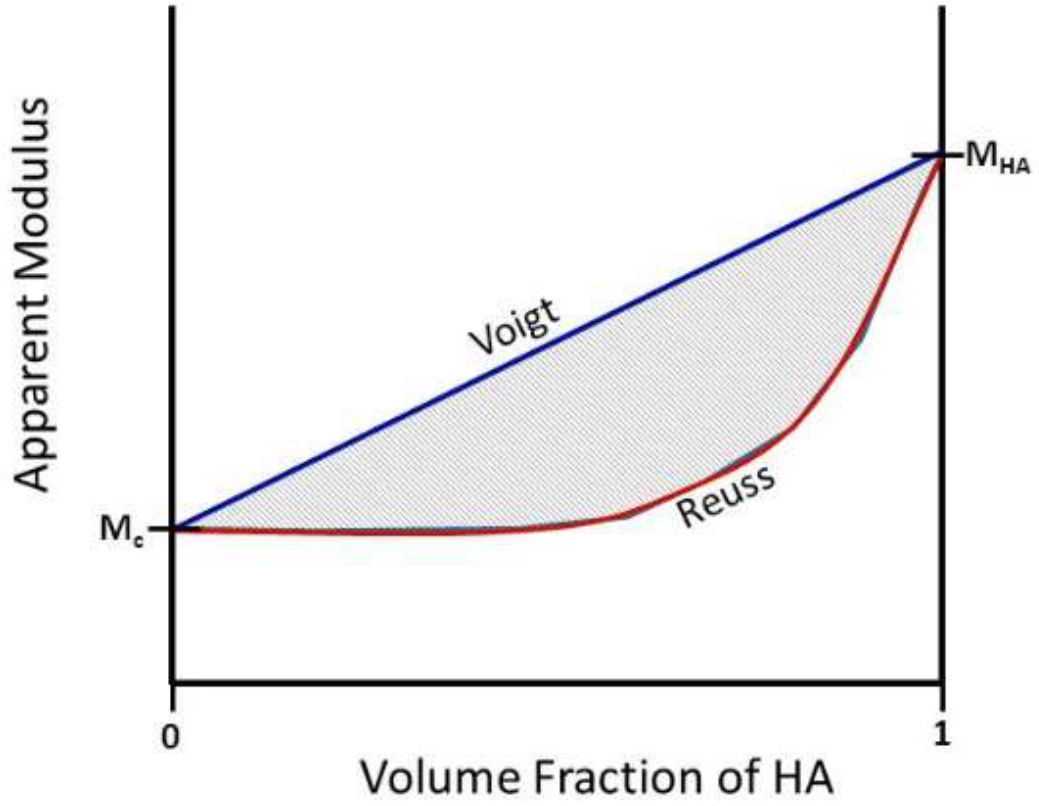
Model as serial components in mixture



$$\frac{1}{M_v} = \sum_{i=1}^{N} \frac{f_i}{M_i}$$

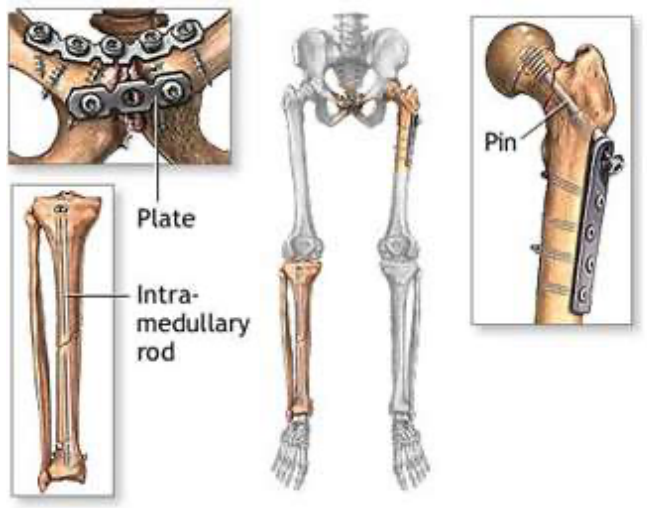
M_i = *Modulus of* i^{th} *constituent*

f_i = *volume fraction of* i^{th} *constituent*



Bone Repair

While the patient is pain-free (general or local anesthesia), an incision is made over the fractured bone. The bone is placed in proper position and screws, pins, or plates are attached to or in the bone temporarily or permanently. Any disrupted blood vessels are tied off or burned (cauterized). If examination of the fracture shows that a quantity of bone has been lost as a result of the fracture, especially if there is a gap between the broken bone ends, the surgeon may decide that a bone graft is essential to avoid delayed healing.

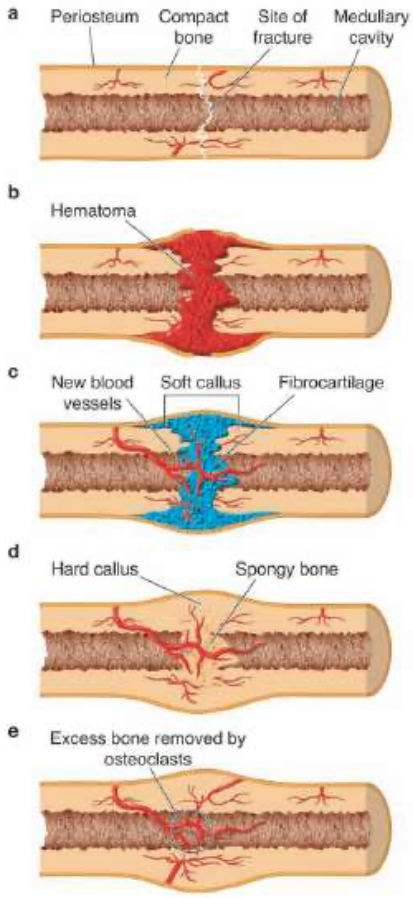


If bone grafting is not necessary, the fracture can be repaired by the following methods:

- one or more screws inserted across the break to hold it.
- a steel plate held by screws drilled into the bone.
- a long fluted metal pin with holes in it, is driven down the shaft of the bone from one end, with screws then passed through the bone and through a hole in the pin.

Repair of a fractured bone

An illustration of the repair of a fractured bone (a) is shown in this diagram. Blood infiltrates the damaged site, forming a hematoma (b), a soft callus of fibrocartilage forms around the hematoma to provide support (c), osteoblasts produce a hard callus to strengthen the soft callus (d), and finally, osteoclasts remove excess bone and callus (e).



From Bone Biology to Biomineralization

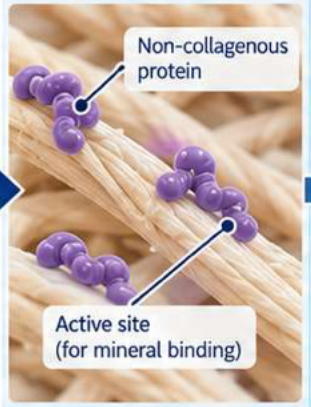
Biomedical Materials — Week 2

- Bone = mineral + collagen + water
- Non-collagenous proteins aid binding
- Minerals are not directly bound to collagen
- Biomineralization starts after matrix deposition
- Biology inspires graft design

1 Collagen matrix laid down

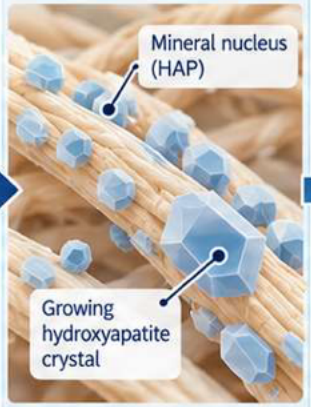
Osteoblasts secrete collagen type I, forming a fibrillar matrix.

2 Non-collagenous proteins provide active sites



Non-collagenous proteins (e.g., osteopontin, osteocalcin) bind to collagen and provide nucleation sites for minerals.

3 Mineral nucleation and growth



Calcium and phosphate ions nucleate at active sites and grow into nanocrystals of hydroxyapatite.

4 Mineralized collagen bone matrix

Collagen fibrils reinforced with aligned hydroxyapatite nanocrystals, forming a mineralized bone matrix.

Learning focus: Composition | Cells | Hierarchy

Next: mechanism of biological mineralization

Elastic Modulus and Stress Shielding

Why implant stiffness matters

Cortical bone
Elastic modulus roughly in the 10–20 GPa range.

Titanium alloys
Lower stiffness than stainless steel and Co–Cr, but still much stiffer than cortical bone.

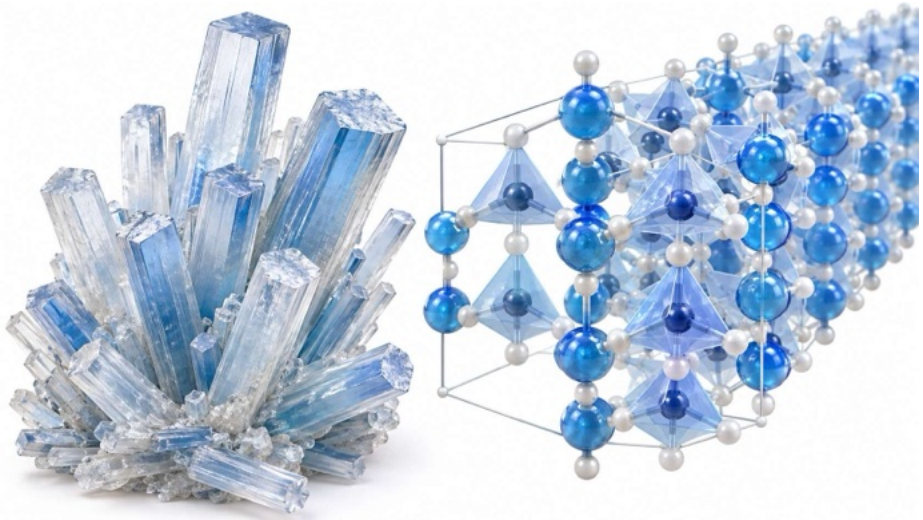
Co–Cr / stainless steels
High modulus, high strength, and high stiffness for load-bearing implants.

Stress shielding
If an implant is much stiffer than bone, it can carry too much load and reduce physiological loading of the surrounding bone.

Design goal: enough stiffness for function, but not so much that surrounding bone is mechanically unloaded.

Hydroxyapatite as a Biomaterial

Why this ceramic appears repeatedly in bone applications



Chemically similar to bone mineral

This similarity helps explain its use in bone-contacting applications.

Can be dense or porous

Dense ceramics resist wear; porous forms support tissue ingrowth.

Often combined with other materials

HA may be used as coatings, granules, blocks, composites, or scaffold phases.

Limitation

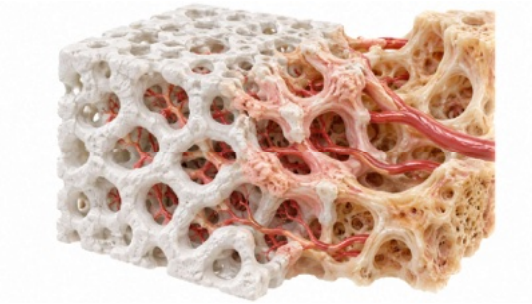
HA is brittle compared with metals and cannot replace every load-bearing function alone.

Dense vs. Porous Hydroxyapatite

Processing changes both structure and function

Dense HA
- lower porosity
- higher stiffness
- easier machining into blocks or granules
- limited space for ingrowth

Porous HA
- interconnected pores
- lower stiffness
- supports tissue and vessel ingrowth
- useful as graft/scaffold-like material

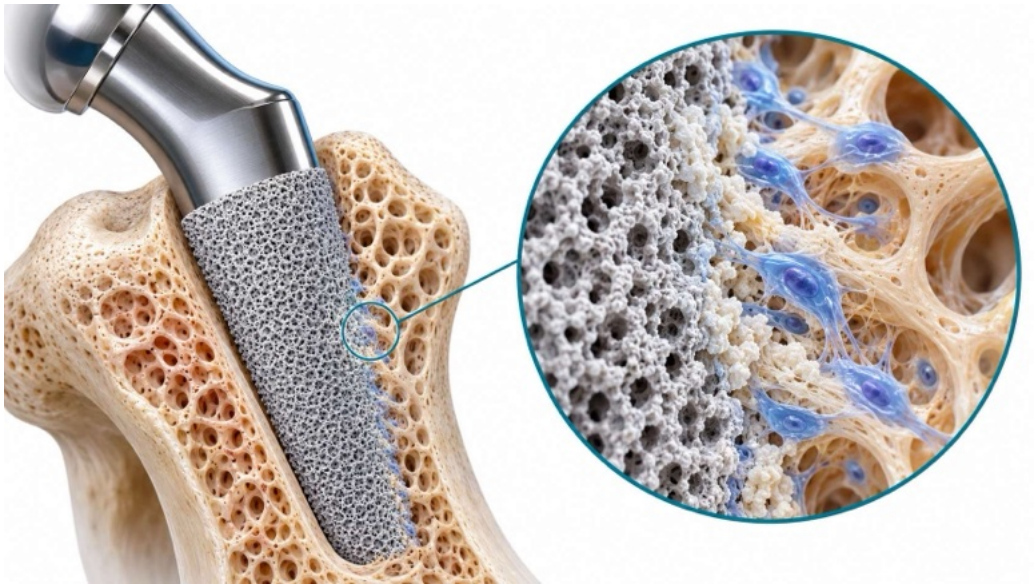


Engineering balance: porosity improves biological access but usually reduces mechanical strength.

Hydroxyapatite Coatings

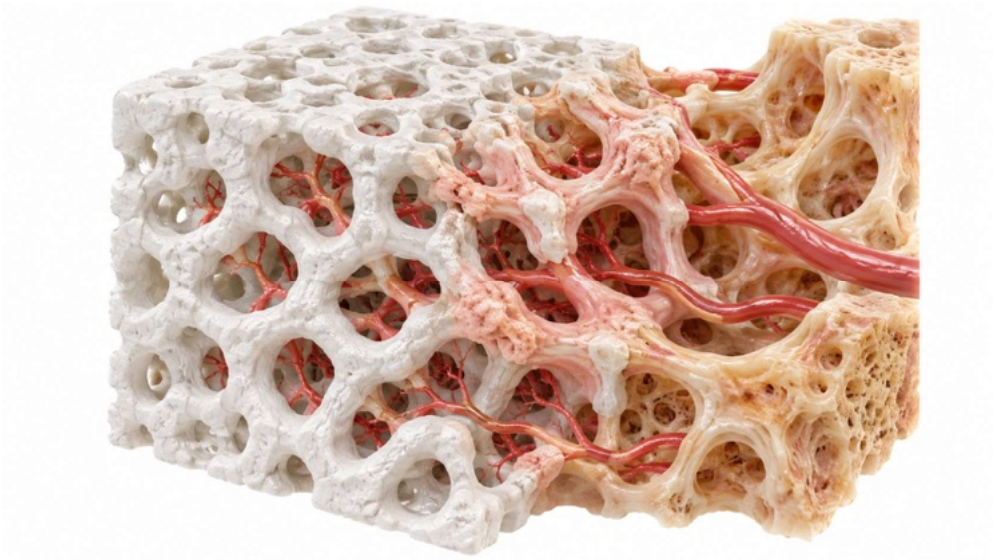
A surface strategy for bone-contacting metal implants

- A metallic implant may provide the load-bearing core.
- An HA coating can encourage direct apposition with bone.
- The coating is only a surface layer, so its adhesion and stability matter.
- Thermal spray routes can form a layered coating structure.
- Coating morphology influences dissolution and biological performance.



Bone Ingrowth and Scaffold Requirements

Porous materials must be designed for biology and mechanics



Interconnected pores
Allow tissue and fluid access throughout the scaffold.

Appropriate pore size
Pores must be large enough for cells, vessels, and new matrix.

Surface chemistry
Should support attachment and mineralized tissue formation.

Mechanical support
Must maintain function during healing or replacement.

Osseointegration and Osteoconduction

Related but not identical concepts

Osseointegration

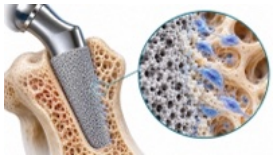
A direct structural and functional connection between bone and implant.

No intervening fibrous tissue layer is desired.

Osteoconduction

A material supports new bone growth by acting as a scaffold or framework for bone-forming cells.

Simple sequence: osteoconductive surface → osteoblast attachment → new bone matrix → stable bone–implant connection.



Bioactive vs. Bioresorbable Bone Materials

Different design intentions for bone repair

Bioactive
Designed to interact positively with surrounding tissue, often forming or stimulating bonding at the interface.

Bioresorbable / biodegradable
Designed to dissolve or degrade and gradually be replaced by advancing tissue.

Example: HA coating
Aim: improve early fixation and bone apposition while metal carries load.

Example: calcium phosphate graft
Aim: act as a temporary bone-like phase during healing.

The correct choice depends on whether we need long-term stability, controlled resorption, or active bone bonding.

Concept Check

Quick questions for student understanding

1. What are the two main architectural types of bone?
2. Why is bone considered a composite material?
3. What does hydroxyapatite contribute to bone?
4. Why does porosity help bone ingrowth but weaken a scaffold?
5. What is the difference between osseointegration and osteoconduction?

Compressive Stress in a Tibia

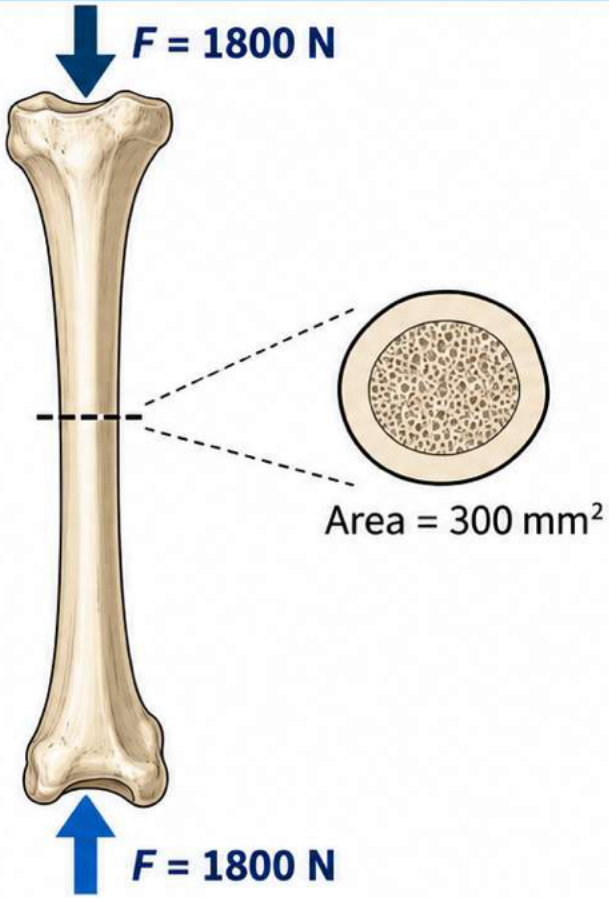
A simplified tibial section is subjected to a compressive load of **1800 N**. The effective load-bearing cross-sectional area of the tibia is **300 mm^2**.

Calculate the compressive stress in the tibia in **MPa**.

Follow-up: If bone loss reduces the effective cross-sectional area while the applied load remains unchanged, what happens to the compressive stress?

Problem 1: Compressive Stress in a Tibia

Question slide

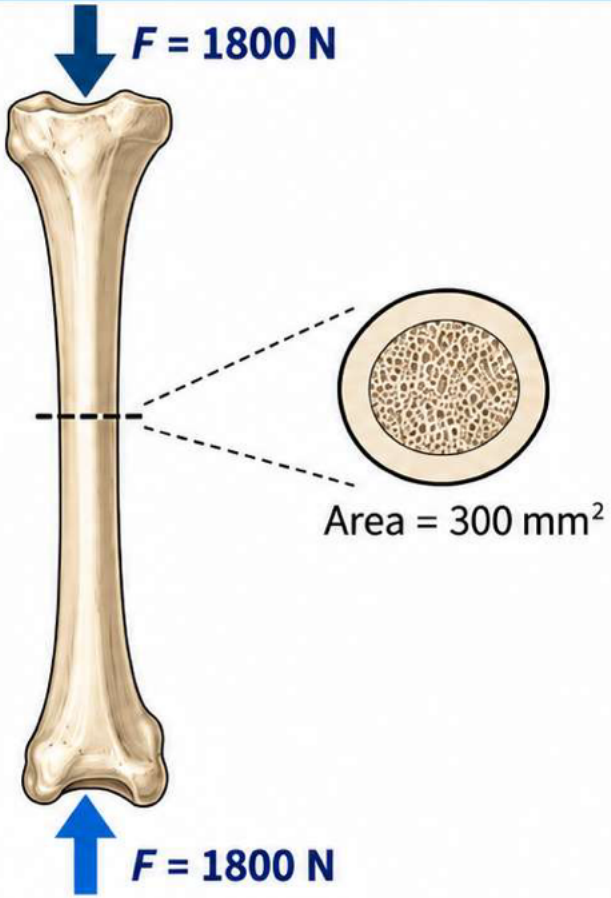


- **Simplified** tibial section
- **Compressive load:** 1800 N
- **Cross-sectional area:** 300 mm^2
- **Find:** compressive stress (MPa)
- **Use:**
- **Think:** effect of bone loss on stress

Work it out before seeing the solution.

Problem 1: Compressive Stress in a Tibia

Solution slide



- **Given:** $F = 1800$ N
- **Given:** $A = 300$ mm²
- **Use:** $\sigma = F / A$
- $\sigma = 1800 / 300$
- $\sigma = 6\ N/mm^2$
- $1\ N/mm^2 = 1$ MPa
- **Answer:** $\sigma = 6$ MPa
- If area decreases, stress increases.

Compressive stress = 6 MPa

Compressive Strain in a Femur

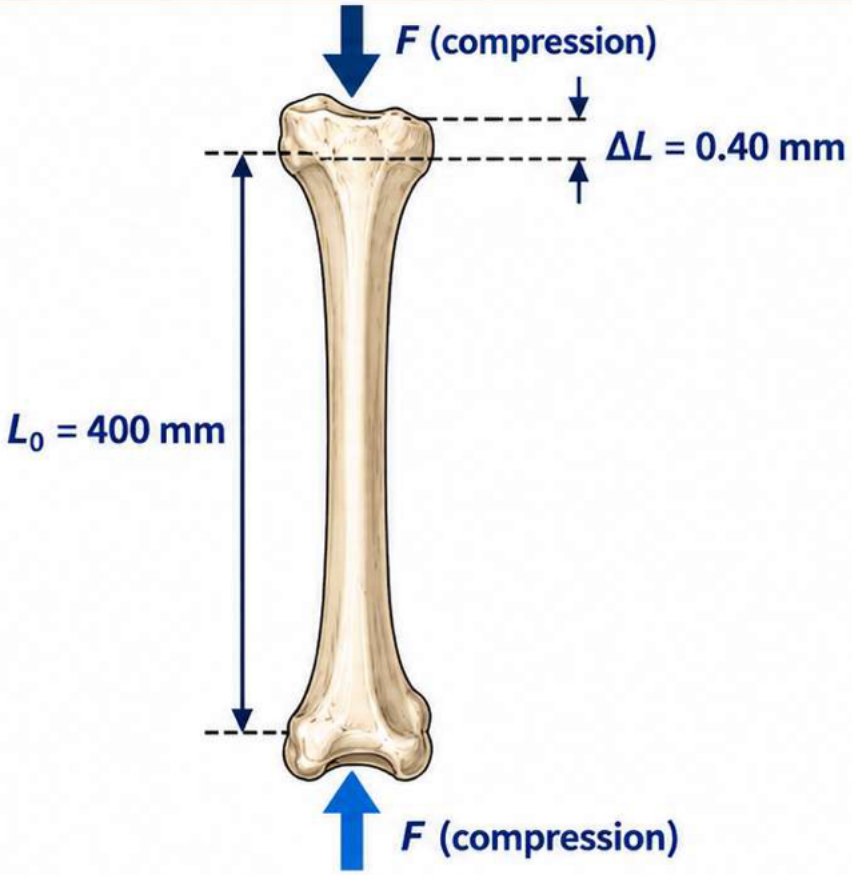
A simplified femur has an original length of **400 mm**. Under a compressive load, the femur shortens by **0.40 mm**.

Calculate:
a) the compressive strain, and
b) the strain expressed as a percentage.

Follow-up: Why is strain a dimensionless quantity?

Problem 2: Strain in a Femur

Question slide

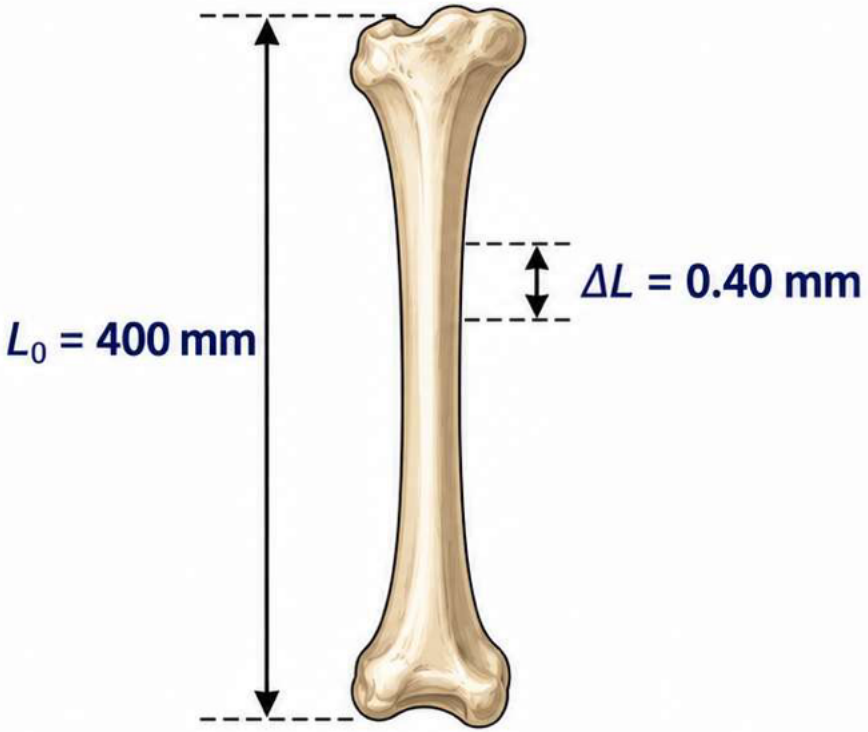


- **Simplified** femur
- **Original length:** 400 mm
- **Shortening:** 0.40 mm
- **Find:** compressive strain
- **Also find:** strain (%)
- **Use:**
- **Think:** why is strain unitless?

Try it before checking the answer.

Problem 2: Strain in a Femur

Solution slide



- **Given:** $\Delta L = 0.40$ mm
- **Given:** $L_0 = 400$ mm
- **Use:** $\varepsilon = \Delta L / L_0$
- $\varepsilon = 0.40 / 400$
- $\varepsilon = 0.001$
- **Strain (%)** $= 0.001 \times 100$
- **Answer:** 0.10%
- Strain has no units.

Compressive strain = 0.001 = 0.10%

Young's Modulus of Cortical Bone

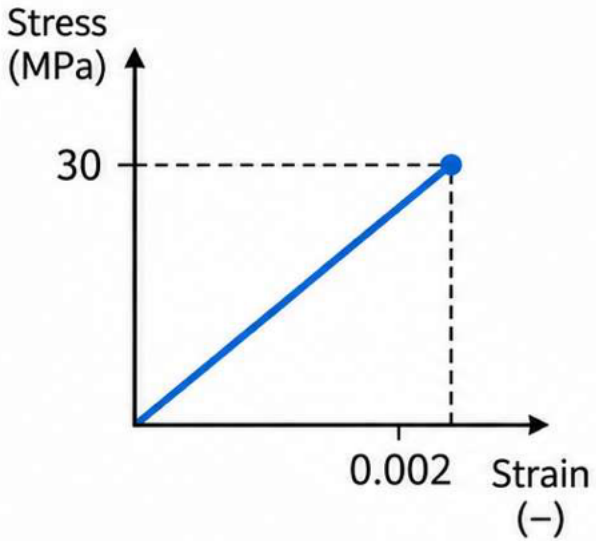
During a compression test on a cortical-bone specimen, the material experiences a stress of **30 MPa** and an elastic strain of **0.002**. Assume that the bone behaves as a linear-elastic material within this range.

Calculate the Young's modulus of the cortical bone and report the answer in **GPa**.

Follow-up: If another bone specimen has a higher Young's modulus, would it deform more or less under the same applied stress?

Problem 3: Young's Modulus of Cortical Bone

Question slide

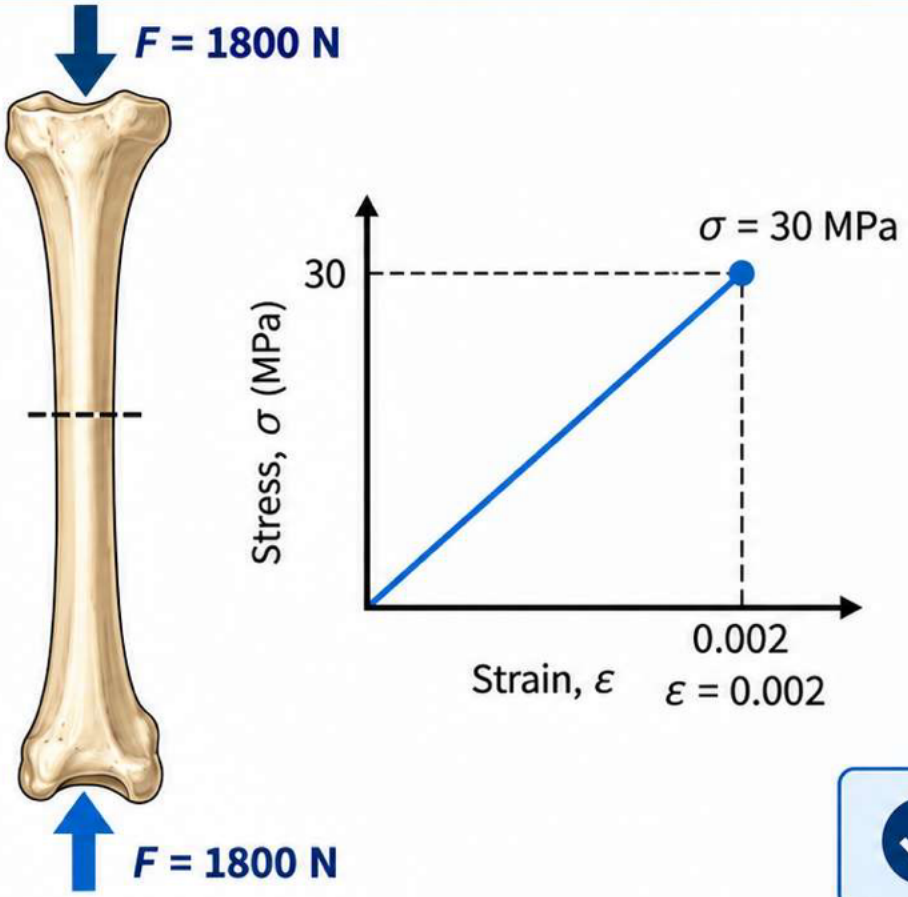


- **Compression** test
- **Stress:** 30 MPa
- **Elastic strain:** 0.002
- **Assume:** linear elastic
- **Find:** Young's modulus
- **Report:** in GPa
- **Use:**

Remember: 1000 MPa = 1 GPa

Problem 3: Young's Modulus of Cortical Bone

Solution slide



- **Given:** $\sigma = 30$ MPa
- **Given:** $\varepsilon = 0.002$
- **Use:** $E = \sigma / \varepsilon$
- $E = 30 / 0.002$
- $E = 15000$ MPa
- **Convert to GPa**
- **Answer:** $E = 15$ GPa
- **Higher** E = stiffer material

Young's modulus = 15 GPa

Bending Stress in a Long Bone

A section of a long bone is simplified as a rectangular beam with a width of **20 mm** and a height of **30 mm**. During loading, the bone experiences a bending moment of **60 N·m**.

Calculate the maximum bending stress in the bone.

Follow-up: During bending, which side of the bone is subjected to tension and which side is subjected to compression?

Bending Stress Formulae for Common Shapes

Useful sections for simplified bone mechanics problems

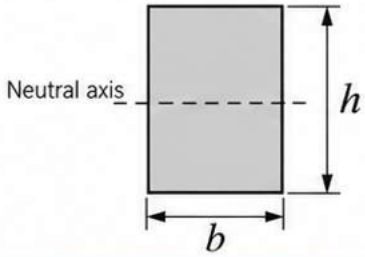
General formula

$$\sigma = \frac{My}{I}$$

At the outer surface: $\sigma_{max} = \frac{Mc}{I} = \frac{M}{S}$

M = bending moment
y = distance from neutral axis
I = second moment of area
c = outer-fiber distance
S = section modulus

A. Rectangular section

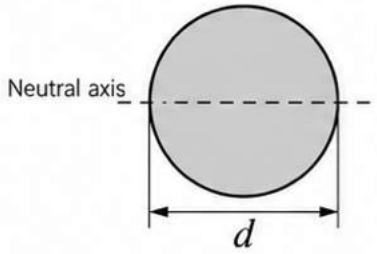


$$I = \frac{bh^3}{12}$$

$$c = \frac{h}{2}$$

$$\sigma_{max} = \frac{6M}{bh^2}$$

B. Solid circular section

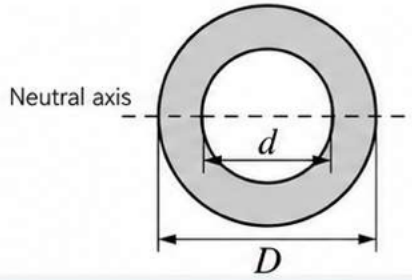


$$I = \frac{\pi d^4}{64}$$

$$c = \frac{d}{2}$$

$$\sigma_{max} = \frac{32M}{\pi d^3}$$

C. Hollow circular section (tube)

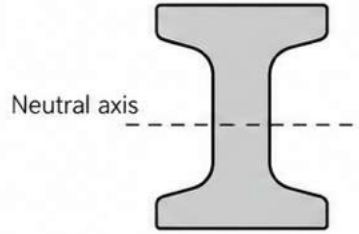


$$I = \frac{\pi(D^4 - d^4)}{64}$$

$$c = \frac{D}{2}$$

$$\sigma_{max} = \frac{32MD}{\pi(D^4 - d^4)}$$

D. Generic symmetric section



$$\sigma_{max} = \frac{M}{S}$$

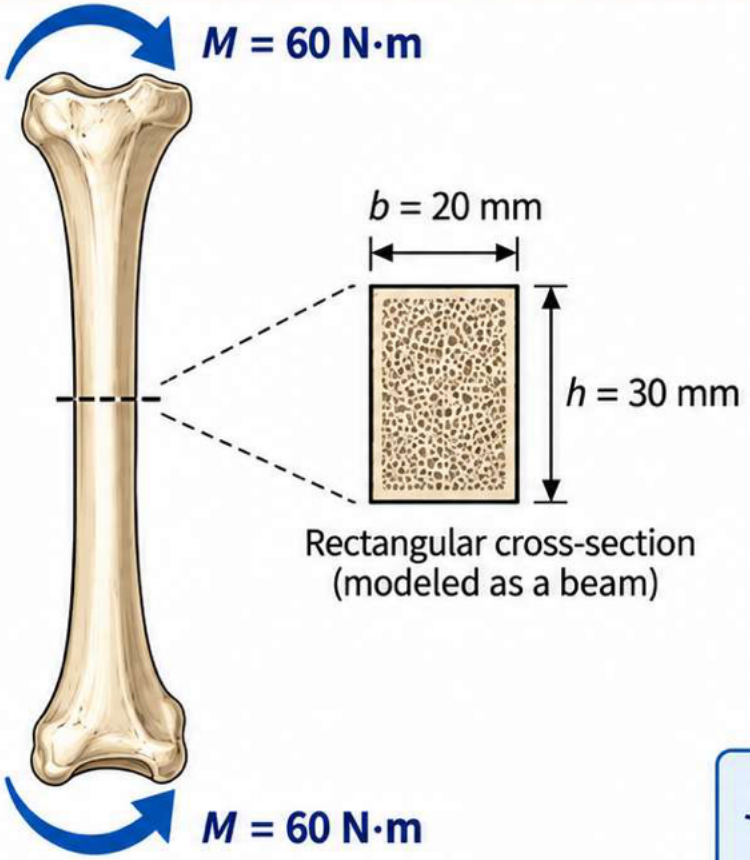
where $S = \frac{I}{c}$

Use this form when I and S are known from tables or CAD.

Bone mechanics tip: Long bones are often simplified as hollow circular or rectangular sections when learning bending basics.

Problem 4: Bending of a Long Bone

Question slide

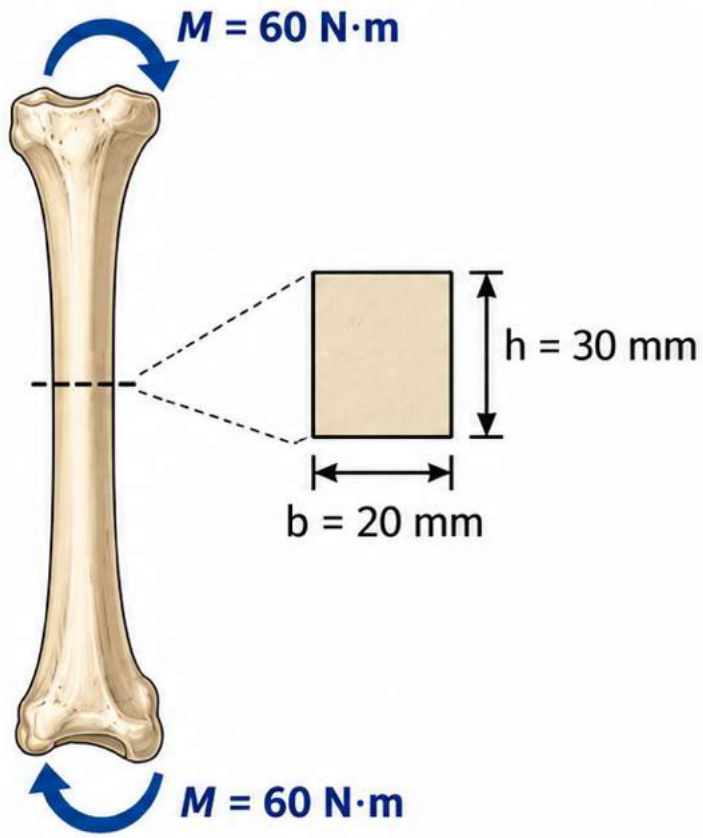


- **Bone modeled** as beam
- **Width *b*:** 20 mm
- **Height *h*:** 30 mm
- **Bending moment:** 60 N·m
- **Find:** maximum bending stress
- **Use:** [redacted]
- **Convert:** *N*·m to *N*·mm

Think about tension and compression sides.

Problem 4: Bending of a Long Bone

Solution slide



- **Convert:** 60 N·m = 60000 N·mm
- **Use:** $\sigma_{max} = \frac{6M}{bh^2}$
- $\sigma_{max} = \frac{6 \times 60000}{20 \times 30^2}$
- $\sigma_{max} = \frac{360000}{18000}$
- $\sigma_{max} = 20\ \text{N/mm}^2$
- $1\ \text{N/mm}^2 = 1\ \text{MPa}$
- **Answer:** $\sigma_{max} = 20$ MPa
- Outer surface sees max stress.

Maximum bending stress = 20 MPa

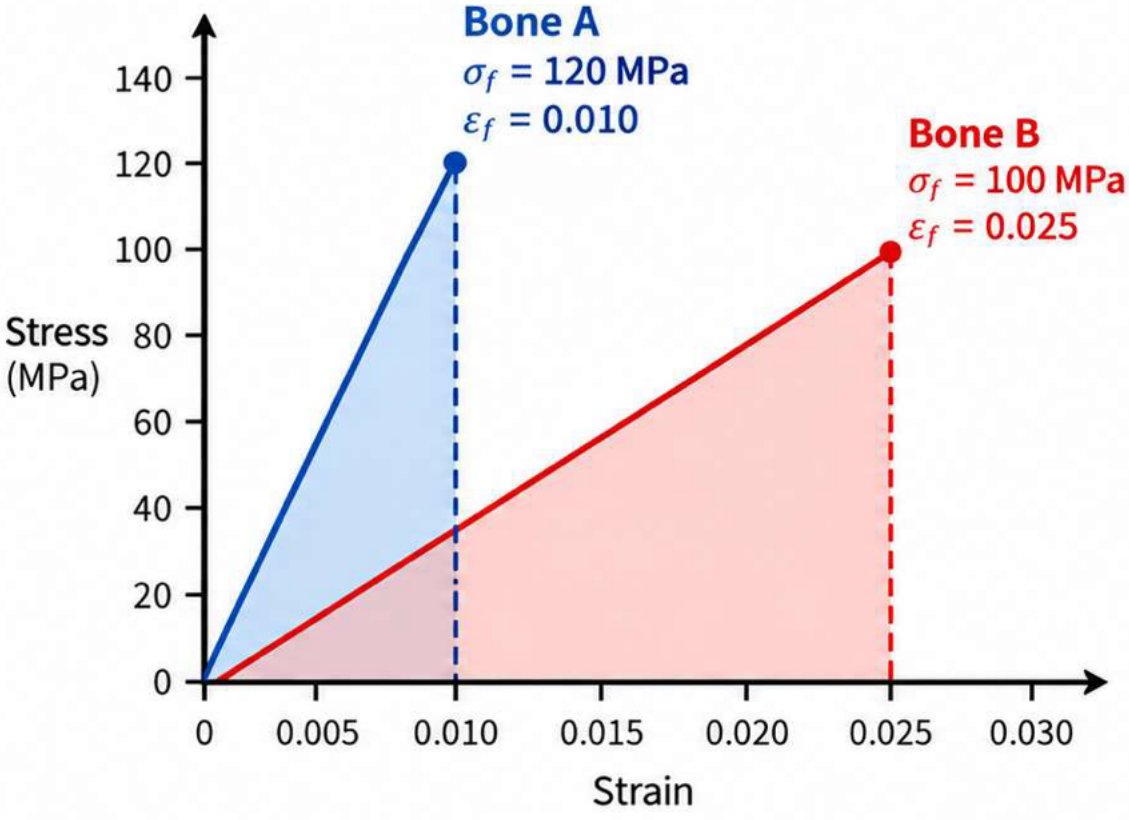
Strength versus Toughness of Bone

Two simplified bone specimens, Bone A and Bone B, are tested until fracture.
For **Bone A**: Stress=120 Mpa, Strain=0.010
For **Bone B**: Stress=100 Mpa, Strain=0.025
Assume that the stress–strain response can be approximated as a straight line from the origin to fracture.

Estimate the toughness of each bone.
Then determine:
a) Which bone is stronger?
b) Which bone is tougher

Problem 5: Strength vs Toughness

Question slide

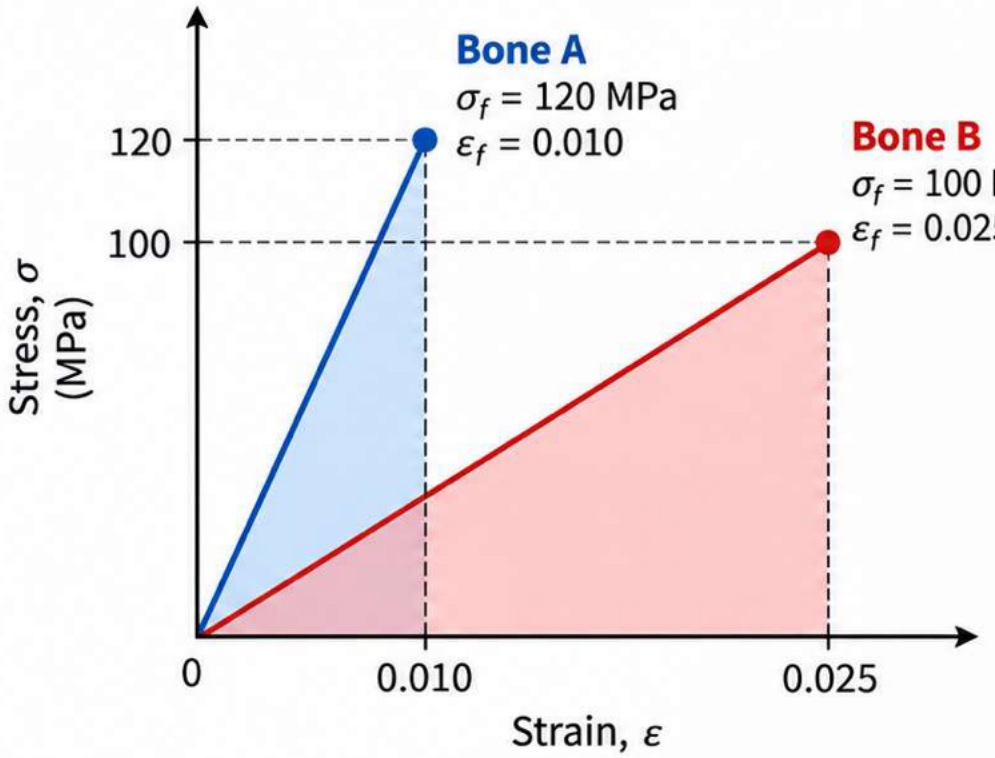


- **Bone A:** $\sigma_f = 120$ MPa
- **Bone A:** $\varepsilon_f = 0.010$
- **Bone B:** $\sigma_f = 100$ MPa
- **Bone B:** $\varepsilon_f = 0.025$
- **Approximate toughness**
- **Use:** [redacted]
- **Find:** toughness of A and B
- **Then decide:** stronger? tougher?

Area under curve relates to toughness.

Problem 5: Strength vs Toughness

Solution slide



- **Use:** $U_T = \frac{1}{2}\,\sigma_f\,\varepsilon_f$
- **Bone A:** $\frac{1}{2} \times 120 \times 0.010$
- $U_{T,A} = 0.60$ MJ/m³
- **Bone B:** $\frac{1}{2} \times 100 \times 0.025$
- $U_{T,B} = 1.25$ MJ/m³
- **Stronger:** Bone A
- **Tougher:** Bone B
- **Reason:** larger area under curve

Bone A is stronger; Bone B is tougher.

Eccentric Compression in a Long Bone: Axial Stress + Bending

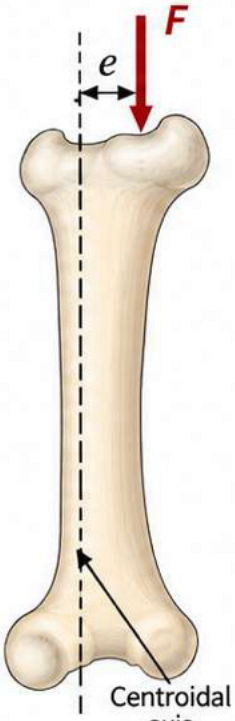
A long-bone section is simplified as a rectangular beam with a width of **25 mm** and a height of **35 mm**. A compressive force of **1800 N** acts at an eccentricity of **12 mm** from the centroidal axis of the section.
Calculate:
a) the cross-sectional area,
b) the axial compressive stress,
c) the bending moment created by the eccentric load,
d) the maximum bending stress,
e) the maximum compressive stress on one surface, and
f) the stress on the opposite surface.

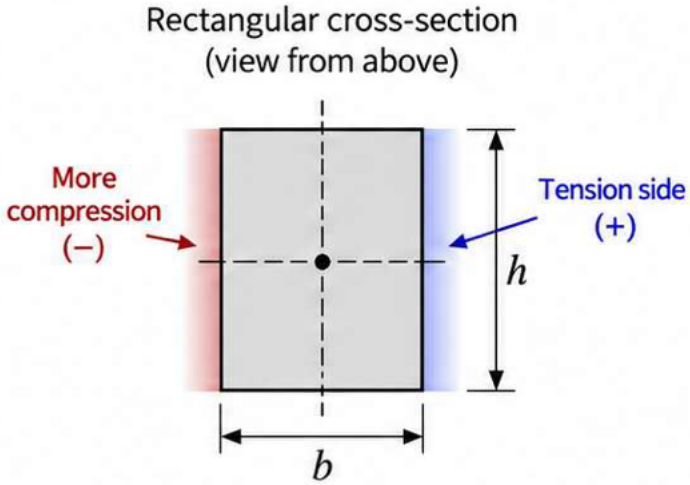
Problem : Eccentric Compression in a Long Bone

Combined axial stress + bending stress

Problem statement:

- Model a long-bone section as a rectangular beam.
- Width $b = 25$ mm
- Height $h = 35$ mm
- Compressive force $F = 1800$ N
- Eccentricity $e = 12$ mm from centroid
- **Find:** axial stress, bending moment, maximum compressive stress, and stress on the opposite surface.





Given data:

$b = 25$ mm $\quad h = 35$ mm $\quad F = 1800$ N $\quad e = 12$ mm

Required formulas:

$$\sigma_a = \frac{F}{A} \qquad M = Fe \qquad \sigma_b = \frac{6M}{bh^2} \qquad \sigma_{max} = \sigma_a + \sigma_b$$

Problem 1 Solution: Eccentric Compression

Complex bone-mechanics solution

Solution steps

Given: b = 25 mm, h = 35 mm, F = 1800 N, e = 12 mm

$A = b\,h = 25 \times 35 = 875\ mm^2$

$\sigma_a = F / A = 1800 / 875 = 2.06\ MPa$

$M = F\,e = 1800 \times 12 = 21{,}600\ N\cdot mm = 21.6\ N\cdot m$

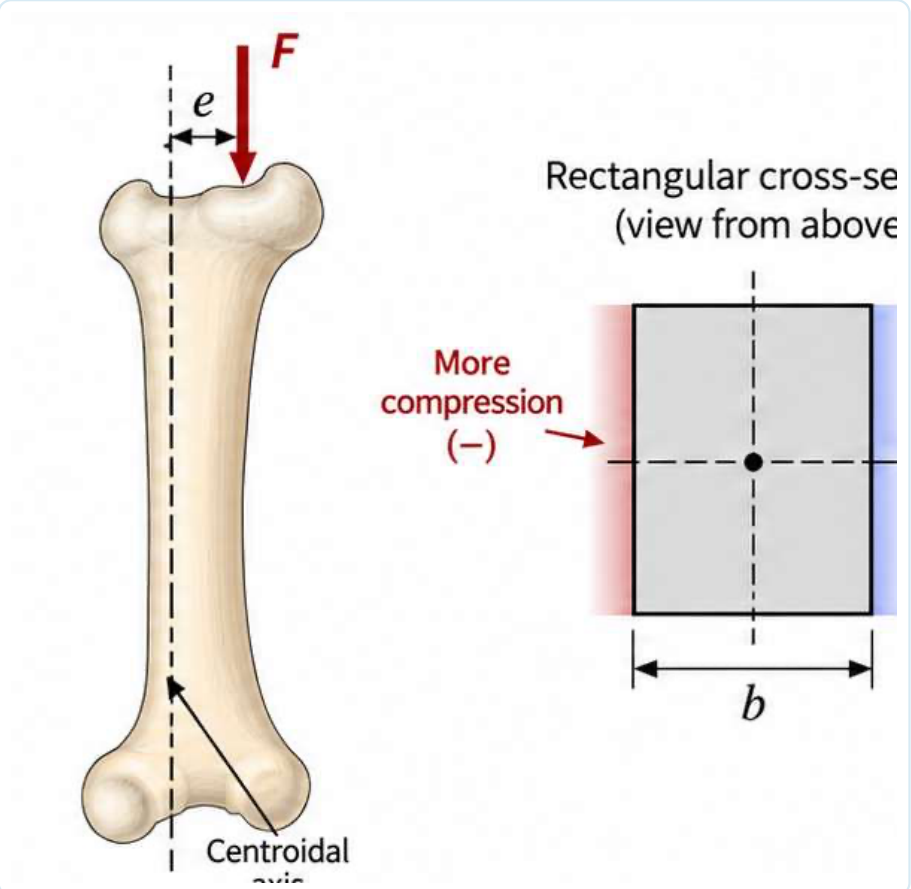
$\sigma_\beta = 6M / (b\,h^2) = 6(21{,}600) / [25(35^2)]$

$\sigma_\beta = 4.23\ MPa$

Maximum compression $= \sigma_a + \sigma_\beta = 2.06 + 4.23 = 6.29\ MPa$

Opposite surface $= \sigma_a - \sigma_\beta = 2.06 - 4.23 = -2.17\ MPa$

Answer: max compression = 6.29 MPa; opposite surface = 2.17 MPa tension



Off-center compression adds bending; one side can become tensile.

Stress, Strain, Shortening and Elastic Energy in Cortical Bone

A cortical-bone specimen has a length of **120 mm** and a cross-sectional area of **400 mm^2**. It is subjected to a compressive load of **12 kN**. Assume the Young's modulus of the bone is **17 GPa** and that the material behaves linearly elastically.

Calculate:
a) the compressive stress,
b) the compressive strain,
c) the shortening of the bone specimen, and
d) the elastic strain-energy density stored in the bone.